



**Verified Carbon
Standard**

VALIDATION REPORT OF UMUT-II HYDROPOWER PLANT, TURKEY

Earthood

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Client	RHG Enertürk Enerji Üretim ve Ticaret A.Ş.
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Summary:

Earthood Services Limited (hereafter referred to as Earthood) has been contracted to validate the project activity " **Umut-II Hydropower Plant, Turkey** " (VCS ID- 1011). The project proponent is **RHG Enertürk Enerji Üretim ve Ticaret A.Ş.** and GTE Karbon Sürdürülebilir Enerji Eğitim Danışmanlık Ve Ticaret A.Ş serves as the project consultant, ensuring compliance with the applicable requirements outlined in the VCS program guidelines and standards (VCS Standard Version 4.7 and VCS Program Guide Version 4.4/06/07/). The project uses the VCS methodology, ACM0002 (Version 22.0)/04/. The project corresponds to the generation of clean electricity from hydropower plant and supplying the electrical power to the grid of Türkiye.

The objective of the validation is to perform a complete and independent review of the proposed project activity against the applicable VCS requirements, including the baseline, monitoring plan, and compliance with the relevant VCS and host country criteria. These are validated to ensure that the project design, as documented, is sound and reasonable, and that it fits the established standards. Validation is a prerequisite for all the VCS projects, and it is viewed as vital to give stakeholders assurance about the project's quality and expected emission reductions. The VVB's aim is to perform a complete, independent evaluation of the project activity.

The scope of the renewal of crediting period of the project validation is described as an objective and independent review of the project description (PD). The VCS PD/02/ is compared to the relevant criteria and guidance documents provided by VCS, including the VCS Program Guide/07/, VCS Standard/06/, Program Definitions/15/, and Issuance Process/23/ in order to ensure that the project currently meets the applicability conditions of the chosen baseline and monitoring methodology, ACM0002: Grid-connected electricity generation from renewable sources /04/.

The scope of the revalidation is to establish that:

- The proposed project activity complies with the corresponding registered VCS-PD and the latest version of the renewed VCS-PD.

- The Proposed project activity complies with all relevant VCS rules and requirements. The VCS renewal of crediting period was based on the updated VCS-PD, estimated emission reduction calculation spreadsheet, information from current validated VCS-PD and supporting documents made available to Earthood by the project participants.
- The proposed monitoring plan complies with all requirements of applied methodology.

The validation, which was conducted following Earthood's internal quality procedures, consisted of the following phases:

- a) Document review, involving:
 - A review of data and information.
 - Cross checks between information provided in the VCS PD/02/ and information from sources other than those used, if available, and if necessary independent background investigations.
- b) Follow-up actions (e.g., site audit and telephone or email interviews), including:
 - Interviews with relevant stakeholders in the host country, personnel with knowledge of the project design and implementation.
 - Cross checks between information provided by interviewed personnel (i.e., by checking sources or other interviews) to ensure that no relevant information has been omitted.
- c) Reference to available information relating to projects or technologies similar to the proposed project activity under validation of renewal of crediting period (RCP).
- d) Review, based on the approved methodology being applied, of the appropriateness of formulae and accuracy of calculations.

A risk-based approach has been followed to perform this validation of RCP. During the validation process, a total of 09 findings were raised, which includes 06 Corrective Action Requests (CARs), 02 Clarification Requests (CLs) and 01 Forward Action Requests (FARs). The project proponent has successfully closed all the issues raised.

The VCS PD/02/, emissions reduction calculations/03/, and accompanying documents provided are all in compliance with VCS criteria. The validation of the RCP was completed with a reasonable level of assurance and no uncertainties were found related to the project activity validation.

Earthood Services Limited (hereafter referred to as Earthood) certifies that the project meets the requirements of the VCS PD template, v4.4, the VCS Standard, v4.7/06/, and the applied methodology: ACM0002: Grid-connected electricity generation from renewable sources /04/. The estimated annual average and total emission reductions from the project activity, for the crediting period 21/02/2024 to 20/02/2034 (both days inclusive) are expected to be 330,130 tCO₂e average annually and 330,130 tCO₂e over the crediting period and has been successfully validated under VCS. Earthood issues a positive validation opinion, certifying that the project meets the applicable VCS standards and thereby proposing that it be registered.

CONTENTS

1	INTRODUCTION	5
1.1	Objective	5
1.2	Scope and Criteria	5
1.3	Reasonableness of Assumptions	6
1.4	Summary Description of the Project	7
2	VALIDATION PROCESS	8
2.1	Method and Criteria	8
2.2	Document Review	9
2.3	Interviews	9
2.4	Site Visits	11
2.5	Resolution of Findings	11
3	VALIDATION FINDINGS	12
3.1	Project Details	12
3.2	Safeguards and Stakeholder Engagement	19
3.3	Application of Methodology	32
3.4	Non-Permanence Risk Analysis	56
4	VALIDATION OPINION	56
4.1	Validation Summary	56
4.2	Validation Conclusion	57
	APPENDIX 1: COMMERCIAL SENSITIVE INFORMATION	61
	APPENDIX 2: DOCUMENTS REFERRED	62
	APPENDIX 3: ABBREVIATIONS	65
	APPENDIX 4: FINDINGS OVERVIEW	66
	APPENDIX 5: COMPETENCY STATEMENTS	82

1 INTRODUCTION

1.1 Objective

Earthood Services Limited has been contracted to carry out validation of the renewal of crediting period for the VCS project activity “Umut-II Hydropower Plant, Turkey” against the relevant requirements of VCS programme guidelines and standard (VCS Standard Version 4.7/06/ & VCS Program Guide Version 4.4/07/). The title of the project activity is “Umut-II Hydropower Plant, Turkey” and its VCS ID is 1011. The project activity is located in the Black Sea region of Türkiye, near Akkus town, Ordu Province, Karakuş River.

The purpose of this validation of Renewal of Crediting period is to have an independent third-party assessment of whether the project activity conforms to the qualification criteria set out in the VCS Standard, version 4.7/06/ to attain real, measurable, additional, and permanent emission reductions. The validation statement opinion is a written assurance that:

- The project complies with all the applicable VCS requirements and can generate the emission reductions stated over the project’s second crediting period.
- The validation of RCP followed the requirements of the current version of the VCS Standard, version 4.7/06/ and VCS Program Guide, version 4.4/07/ to ensure the quality and consistency of the validation work and the report.
- The project shall result in emission reductions as declared by the organisation or GHG project’s GHG assertion.

In order to fulfil the above objectives, the VVB assigned the task of assessment to validation team that is collectively qualified as per Earthood’s QMS manual and other internal requirements. The data reported is accurate, complete, consistent, transparent, and free of material error or omission.

1.2 Scope and Criteria

The scope of the validation of the renewal of crediting period is to establish/verify that:

- The project activity is in accordance with rules and legislation of the host country (Democratic Republic of Türkiye).
- The project activity is in accordance with all relevant VCS rules and requirements, which includes the following: VCS Program Guide v.4.4/07/ and VCS Standard v.4.7/06/, Program Definitions v.4.5/15/, Registration and Issuance Process v.4.6/23/ and VCS Validation and Verification Manual v.3.2/08/, including additionality, proof of title and compliance with local laws.

- The project activity is in accordance with the conditions of the latest version of applied VCS Methodology: ACM0002: Grid-connected electricity generation from renewable sources /04/.
- the validation of the renewal of crediting period is in accordance with requirements of CDM methodological tool “Assessment of the validity of the original / current baseline and update of the baseline at the renewal of the crediting period” – version 3.0.1 /33/.

The validation of the renewal of crediting period is based on the registered VCS-PD, revised VCS-PD and estimated GHG emission reduction calculations.

The validation of Renewal of crediting period of the project activity process is divided into four stages:

1. Desk review of Project Description (PD)
 - Evaluation of data and information
 - Cross-checks between information provided in the VCS PD/02/ and the registered VCS PD/30/ alongside information obtained from other sources using all available resources without regard to the project proponent's information.
2. On-site audit and follow-up interviews with project stakeholders
 - Telephone, email, or direct on-site interviews with relevant stakeholders in the host country with professionals with expertise of the project development.
 - Cross-checking the information provided by interviewed people with the information provided by the project proponent using all available means.
3. Reference to publicly available information about projects or technologies that are similar to the project under validation and review, using the accepted methodology for formulae and calculation accuracy.
4. The final validation report and opinion, as well as the resolution of remaining concerns

1.3 Reasonableness of Assumptions

☐ Limited Level of Assurance

☒ Reasonable Level of Assurance

The approach used by Earthood for validation of the renewal of crediting period is built on a thorough understanding of the risks associated with reporting data on GHG emissions and the controls used to mitigate them. Earthood conducted the validation of RCP by reviewing substantiating evidence and other relevant information and explanations from sources to provide reasonable assurance that estimated GHG emission reductions are fairly reported.

In the draft validation report (prepared by validation team), the information provided is reviewed by an independent technical review team (one or more members) to confirm if the internal procedures established and implemented by Earthood were duly complied with and such opinion/conclusion is reached in an objective manner that complies with the applicable VCS and CDM (Clean Development Mechanism) requirements as appropriate. The technical review team is collectively required to possess technical expertise of all the technical area/sectoral scope the project activity relates to. All team members of the technical review team are independent of the validation team. The report approved by the Technical Manager has been endorsed by the Chief Executive Officer, who is overall responsible for ensuring quality, before final release.

In our opinion the estimated GHG emissions reductions were calculated correctly based on the approved baseline and monitoring methodologies ACM0002: Grid-connected electricity generation from renewable sources /04/. The assessment result has a reasonable level of assurance in validation that GHG assertions are free of material errors, omissions, and misrepresentations. The document reviewed are included under Appendix 2 of this report.

1.4 Summary Description of the Project

The summary description of the project activity has been illustrated in section 1.1 of VCS PD/02/ appropriately. As per the given details in VCS PD, information was assessed on-site and found to be consistent. The project activity "Umut-II Hydropower Plant, Turkey" involves generation of clean electricity from run-of-river hydroelectric power plant. The project is an initiative undertaken voluntarily by its proponent. There are no regulatory constraints pertaining to the Hydro Power project in the host country's laws and regulations. The project proponent is RHG Enertürk Enerji Üretim ve Ticaret A.Ş. and is the legal/project owner of the hydropower plant and GTE Karbon Sürdürülebilir Enerji Eğitim Danışmanlık Ve Ticaret A.Ş. is involved as a project consultant in document preparation for the project activity.

The project activity is located in the Black Sea region of Türkiye, near Akkus town, Ordu Province, Karakuş River. The Powerhouse is located close to the Gokcebayir village, and the diversion weirs are located close to the Karakus village and is expected to generate 72,276 MWh annually/09/, /03/. The Hydropower plant consists of three Francis type turbines with 8.435 MW_m/8.150 MW_e each, totaling to 24.45 MW_e a reservoir with 87,327 m² surface area with the power supply to the grid being 72,276 MWh per annum, which has been checked through the generation license/27/, feasibility study report/16/ and Survey for Area of Reservoir/24/. The VCS Project activity " Umut-II Hydropower Plant, Turkey " is adjacent to the existing Umut-I and Umut-III hydro power plant. They share: diversion weir, conveyance tunnel, forebay, and penstock, along with the same legal owner, and employs the same technology to generate electricity. Umut-I Hydropower Plant, Turkey & Umut-II Hydropower Plant, Turkey & Umut-III Hydropower Plant, Turkey have the same electricity generation license/27/ and connected to the same electricity meters. Although the project proponent presents separate gross electricity generation figures for each plant, these are not based on individual metering. Instead, the total measured output is proportionally allocated among the three plants based on their installed capacities: 5.800 MW (Umut-I), 24.450 MW (Umut-II), and 12.000 MW (Umut-III). Accordingly, a proportional constant of 0.5787 is used to estimate Umut-II HEPP's share of the total generation. EPIAŞ data reflects the combined output of all three plants, which operate under RHG Enertürk Enerji Üretim ve Ticaret A.Ş. Based on the total licensed annual generation of 124,894 MWh, the estimated annual output for Umut-II HEPP

is 72,276 MWh. Clean energy produced from the project activity will be supplied to Türkiye grid which will reduce equivalent amount of anthropogenic emissions emitted from the grid connected fossil fuel-based power plants or by the addition of new plants. The project has been commissioned on 21/02/2014, which has been verified through the Provisional acceptance certificate/09/.

The project activity is going to reduce an estimated amount of 33,013 tCO₂e of emission reductions annually and 330,130 tCO₂e of emission reduction over the entire crediting period i.e. 21/02/2024 to 20/02/2034 (both days inclusive). As per section 3.8 of VCS Standard v.4.7/06/, The project start date of a non-AFOLU project is the date on which the project began generating GHG emission reductions or carbon dioxide removals. The date of provisional acceptance/09/ of Umut-II Hydropower Plant, Turkey as described in VCS PD/02/ is 21/02/2014 which can therefore be termed as start date for the project activity.

2 VALIDATION PROCESS

2.1 Method and Criteria

Earthood has been contracted for the validation of the renewal of crediting period of the project activity “Umut-II Hydropower Plant, Turkey” (VCS ID- 1011), the PP of the project is RHG Enerji Üretim ve Ticaret A.Ş., who is also the legal/project owner, and the project consultant is GTE Karbon Sürdürülebilir Enerji Eđt. Dan. ve Tic. A.Ş. The project activity adheres to the relevant requirements of VCS Standard, Version 4.7/06/. The project activity uses the ACM0002: Grid-connected electricity generation from renewable sources /04/, which is classified as sectoral scope 1 “Energy industries (renewable - / non-renewable sources)”. The following were included in the validation of renewal of Crediting period:

- Contract with PP for the scope and appointment of validation team and technical review team.
- Completeness check of VCS PD/02/.
- Desk review of VCS PD/02/ and estimated GHG emission reduction calculation sheet/03/.
- On-site audit by validation team.
- Follow up interviews with representatives of Project Owner
- Reporting and closure of findings (CARs/CLs/FARs) and preparation of draft validation report
- Independent technical review of the draft validation report and final/ revised documentation (e.g. VCS PD, corresponding estimated ER calculation sheet and evidence)
- Issuance of the final validation report to the contracted

In accordance with the VCS Validation and verification manual/08/, VVBs must confirm that the validation can be completed within the relevant validation deadline, relative to the project start date (ie, the date the project starts generating emission reductions and removals). The start date of operation as cross-verified from the provisional acceptance report /09/ is 21/02/2014.

The on-site audit /13/ for the project activity was conducted by the validation team on 27/06/2024 where the key responsibility of the team was to assess the project status which is to be in accordance with the VCS Requirements/06/07/08/. Key stakeholders were identified independently and interviewed in an unbiased manner (free from the influence of the PP).

Table 1: Validation of RCP Schedule

Key Milestones	Date
Kick-off meeting	01/06/2024
Desk Review	07/06/2024- 27/05/2025
Site Visit	27/06/2024
Draft Validation Report	30/05/2025

2.2 Document Review

A desk review was conducted by the validation team that included:

- A review of the data and information presented to assess its completeness.
- A review of the VCS PD/02/ and registered VCS PD /30/, estimated GHG emission reduction calculation sheet/03/, the applied methodology/04/ including applicable tool(s) and relevant laws regulations for the host country.
- Supporting documents as provided by the Project proponent.

In addition, the validation team consulted third-party documents such as technical reports pertaining to the project design or design or basic condition, and technical data. A complete list of documents/evidence reviewed is included as Appendix 2.

2.3 Interviews

In accordance with para 4.1.11 of the VCS standard v4.7/06/, “A site visit that includes a visit to facilities and/or project areas shall be conducted at validation and project crediting period renewal validation”. The VVB has conducted an onsite audit on 27/06/2024 /13/ to check the implementation status of the project activity. The audit was conducted as a part of validating the project to check the technology employed in the hydropower plant, its location (Physical address),

Monitoring plan (feasibility of monitoring arrangements described in PDD, QA/QC procedures, responsibility of implementation of monitoring plan, data recording & storage procedures). During the Site audit, the representatives of the project proponent were Interviewed on various aspects related to the implementation of the project activity. The questionnaires asked during the on-site audit has been discussed below:

- Introduction, scope and objective of work, roles and responsibilities, resources required, and timetable of the remote audit including closing meeting and any concerns from PP.
- Approval of project activity from Host Party and approval of participation of Project Proponent(s).
- Project boundary and emission sources included in the project boundary.
- All the details about the project activity which are mentioned in the VCS PD were cross-checked by discussion with the PP.

The table below describes the on-site interview process and further identifies personnel, including their roles, who were interviewed and/or provided information additional to that provided in the project description and any supporting document.

S.No.	Name	Organization	Date	Topic	Team
1.	Arzu Meral	Project Expert, GTE Karbon Sürdürülebilir Enerji Eğitim Danışmanlık Ve Ticaret A.Ş.	27/06/2024	Project implementations, start date as per the VCS requirements	Divij Varshney and Kubra Agriman
2.	Saffet Arslan	RHG Enertürk Enerji Üretim ve Ticaret A.Ş.		Available documentation	
3.	Era Kumt	RHG Enertürk Enerji Üretim ve Ticaret A.Ş.		Emission Reduction Ex-ante estimation	
4.	M. Kutalmis Karacan	RHG Enertürk Enerji Üretim ve Ticaret A.Ş.		Reliability, accuracy and assurance of monitoring plan	
5.	Sedat Demir	RHG Enertürk Enerji Üretim ve Ticaret A.Ş.			

6.	Yakup Gam	RHG Enertürk Enerji Üretim ve Ticaret A.Ş.		and practices in operation Calibration procedure Monitoring and measuring system Data collection process Local stakeholder consultation Technical specifications QA/QC procedures, maintenance Practices. Sustainable Development Goals	
7.	Mustafa Erter	Mukhtar (Local Stakeholder)	27/06/2024	Feedback regarding the project, Grievance mechanism, Sustainable development Goals.	Divij Varshney and Kubra Agriman
8.	Behiya Erter	Restaurant owner/ cook (Local Stakeholder)			
9.	Semih Erter	Restaurant Owner (Local Stakeholder)			

2.4 Site Visits

An on-site audit of the Hydro power plant facility was undertaken by the validation team on 27/06/2024/13/. The site audit was conducted to assess the implementation and operation of the project activity and to review evidence, and interview key personnel to confirm evidence associated with the project design, implementation, plant operations, environmental impacts, stakeholders etc.

The details of the interviews conducted, and on-site activities undertaken have been described under section 2.3 above.

2.5 Resolution of Findings

This section represents the conclusions from the validation of the project activity. The results of the document review, remote evaluations, and interviews are presented in this section. CARs,

CLs and FARs are used to correct material inconsistencies discovered during validation of renewal of crediting period.

Corrective action requests (CARs) are issued where:

- i. Mistakes have been made with a direct influence on project results requiring adjustments of the VCS monitoring report.
- ii. Applicable methodological specific requirements have not been met.
- iii. There is a risk that emission reductions cannot be monitored or calculated.

A clarification requests (CL) maybe used where additional information is needed to fully clarify an issue or where the information is not transparent enough to establish whether a requirement is met.

- i. The actual project monitoring and reporting practices requires attention and/ or adjustment for the next consecutive verification period, or
- ii. An adjustment of the monitoring plan/02/ is recommended.

In the context of FARs, risks have been identified, which may endanger the delivery of high-quality emissions reductions in the future, i.e., by deviations from standard procedures as defined by the monitoring plan. Consequently, such aspects should receive a special focus during the next consecutive verification. A FAR may originate from lack of data sustaining claimed emission reductions. FARs does not relate to VCS requirements for registration.

CARs and CLs are to be resolved or closed out if the PP modifies the project description, rectifies the PD or provides adequate additional explanations or evidence that satisfies the concerns. If this is not completed, the project activity cannot be recommended for registration under VCS registry. A total of 09 findings were raised, where 06 CARs, 02 CLs and 01 FARs were raised during the validation. All the findings that are raised and communicated to Project Proponents during the validation are included in Appendix 3.

2.5.1 Forward Action Requests

No Forward Action Requests (FARs) were raised during the validation.

3 VALIDATION FINDINGS

3.1 Project Details

The project activity “Umut-I Hydropower Plant, Turkey” involves generation of clean electricity from a hydroelectric power plant. The Project is located in the Black Sea region of Türkiye, near Akkus town, Ordu Province, Karakuş River with the powerhouse located close to the Gokcebayir village, and the diversion weirs are located close to the Karakus village. The total installed capacity of

24.450 MWe and is expected to generate 72,276 MWh annually. The validation confirms with reasonable level of assurance that the project description as described in section 1.1 of VCS PD/02/ is accurate, complete and therefore provides a complete overview of the project activity. VCS 1011 project meets all applicable rules and requirements which are set out under the VCS Program including VCS Standard Version 4.7/06/ & VCS Program Guide Version 4.4/07/.

Item	Evidence gathering activities, evidence checked, and assessment conclusion				
	Audit type	Period	Program	Validation/verification body name	Number of years
Audit history	Validation	21-February-2014 to 20-February-2024 (1 st Crediting Period)	VCS	TÜV Rheinland Ltd.	Ten years
	Verification	21-February-2014 to 20-February-2020 (1 st Crediting Period 1 st Monitoring Period)	VCS	Earthood Services Limited	Six years
	Verification	21-February-2020 to 20-February-2024 (1 st Crediting Period 2 nd Monitoring Period)	VCS	Re-carbon	Four years

	Renewal of crediting period 21-February-2024 to 20-February-2034 (2 nd Crediting Period)	VCS	Earthood Services Limited	Ten years
	The details of the audit history as described under section 1.2 of the MR and above have been confirmed from the publicly available information and previous verification reports on the project webpage and is found consistent/42/.			
Sectoral scope	Sectoral scope applicable to the Project is: “Type-I: Renewable Energy Projects”			
AFOLU project category, if applicable	Not applicable			
Project activity type	Grid-connected electricity generation from renewable sources			
General eligibility of the project to participate in the VCS Program	The project activity is a 24.450 MW large scale hydro power plant located in Türkiye, which has been verified through the site audit/13/, Feasibility Study report/16/ and provisional acceptance certificate/09/. The project is located in a non-LDC /18/ (Least Developed Country, as designated by the United Nations) and is excluded under Table 2.1 of the VCS Standard version 4.7/06/. However, the project has already been registered under the Verra registry, and the scope of the project activity is Renewal of Crediting Period (RCP). As per para 3.9.8 (4) of the VCS Standard v 4.7/06/, “the project shall be validated against the (current) scope of the VCS Program. Such validation report shall be issued after the end of the (previous) project crediting period and within two years after the end of the (previous) project crediting period.” Consequently, the validation was completed within two years from the end of the previous crediting period, meeting the VCS validation completion requirements as stated in the VCS Standard v4.7/06/. PP has applied large-scale CDM approved methodology, ACM0002 v22.0 /04/, which is found to be eligible under the			

	VCS program as verified from the Verra website /37/. The capacity limit has been verified as per para 119 (a) of CDM PS for PA/38/, where the project is categorized as large-scale with the minimum capacity limit of 15 MW, which is within the limit of installed capacity of project activity, hence, found to be appropriate. The 24.450 MW hydro power plant is not a fragmented or a debundled component part of a larger project as it was cross verified during the on-site audit/13/. The scope of validation is the renewal of the crediting period, and the eligibility of the project activity has already been demonstrated during the registration of the project under VCS Standard v3.0.
AFOLU project eligibility, if applicable	Not applicable as the project activity involves generation of clean energy from operation of 24.450 MW hydroelectric dam.
Transfer project eligibility, if applicable	Not applicable as the project activity is not a transfer project or a CPA seeking registration.
Project design	The Project activity is not a group project activity as it is located in single location. The location of the 24.450 MW hydro power plant was verified during the on-site audit/13/ by the validation team and cross-verified with KML files/11/ provided by PP.
Project ownership	The project activity is developed, commissioned, operated and owned by RHG Enertürk Enerji Üretim ve Ticaret A.Ş. which has been verified from the generation license issued by the EMRA (Energy Market Regulatory Authority)/27/. RHG Enertürk Enerji Üretim ve Ticaret A.Ş is the owner of the carbon credits generated, and GTE Karbon Sürdürülebilir Enerji Eğitim Danışmanlık Ve Ticaret A.Ş is the project consultant, which helps Project Proponent (PP) with the documentation in the registry for the project activity. PP shall be responsible for arranging for verification and issuance of the VCUs generated during the stipulated crediting period. The validation team has verified the details, and it is found to be acceptable.
Project start date	The start date of the Project activity is 21/02/2014, which is date of provisional acceptance of the power plant. The date has been verified using the provisional acceptance certificate /09/. The start date of the Project activity complies with the section 3.8 of the VCS Standard Version 4.7/06/.

Project crediting period	The length of the second (current) crediting period chosen by the PP is ten years, renewal twice in accordance with the para 3.8.1 of the VCS Standard version 3.8. The second crediting period of the project activity is from 21/02/2024 to 20/02/2034, which complies with the section 3.8 of the VCS Standard v3.8. The first crediting period was from 04/10/2012 to 03/10/2022. The scope of validation is the renewal of the crediting period, and the project has been registered under VCS Standard v3.8, where the project crediting period can be a maximum of ten years which may be renewed at most twice. Hence, the chosen crediting period is found to be appropriate.
Project scale	The scale of the project activity is 'Projects' in VCS, which is found to be in line with para 3.10.1 of the VCS Standard/06/, as the annual emission reduction is less than 300,000 tCO _{2e} per year.
Likelihood of achieving estimated GHG emission reduction or removals	The project activity is estimated to generate 33,013 tCO _{2e} emission reduction per year and 330,130 tCO _{2e} over ten years of crediting period. The estimated ERs has been verified through the ER sheet/03/ and is found to be appropriately calculated in line with the applied methodology/04/. The estimation of the electricity generation has been taken from the generation license/27/, which is an official document from the EMRA (governmental entity in host country, Türkiye) and hence, found to be appropriate.
Technologies and measures implemented by the project activity or activities	The project activity "Umut-II Hydropower Plant, Turkey" is operation of 24.450 MWe hydroelectric project located in the Black Sea region of Türkiye, near Akkus town, Ordu Province, Karakuş River which has been checked through the on-site audit/13/, provisional acceptance certificate/09/, and feasibility study report/16/. The project is run-of river project which is owned by RHG Enertürk Enerji Üretim ve Ticaret A.Ş. and will act as the Project Proponent. The Karakuş river is diverted, and water is stored in a reservoir, which has been reviewed through the feasibility study report/16/ and on-site audit/13/. The Project employs 3 Francis-type turbines with 8.150 MWe capacity each, totaling 24.450 MW _e . The project activity is expected to supply a net electricity of 72,276 MWh per annum, which has been verified through the feasibility

	report/16/, generation license/27/, Single line diagram of project/19/ and provisional acceptance certificate/09/. The specifications above were cross checked from technical specification of turbines which have been presented in Feasibility Study Report/16/ submitted by the PP and through the on-site audit/13/.															
Implementation schedule of the project activity or activities	The construction of the project activity was commissioned on 21/02/2014, which has been reviewed through the provisional acceptance certificate /09/ shared by the PP. The project is currently generating renewable electricity which was verified during the on-site audit.															
Project location	The location of the Project activity is Türkiye. The Project is located in the Black Sea region of Türkiye, near Akkus town, Ordu Province, Karakuş River as mentioned in the provisional acceptance report/09/, and feasibility study report/16/. This has also been verified during on-site audit /13/ and cross-verified with KML files/11/ provided by PP. Geographic coordinates provided and are found to be accurate and acceptable. <table><tr><th>Coordinate System</th><th>Diversion Weirs</th><th>Powerhouse</th></tr><tr><td>Degrees, Minutes, Seconds (Latitude)</td><td>40 ° 42' 05"</td><td>40 ° 42' 46"</td></tr><tr><td>Degrees, Minutes, Seconds (Longitude)</td><td>37 ° 4' 12"</td><td>37 ° 1' 44"</td></tr><tr><td>Decimal (Latitude)</td><td>40.701389</td><td>40.712778</td></tr><tr><td>Decimal (Longitude)</td><td>37.07</td><td>37.028889</td></tr></table>	Coordinate System	Diversion Weirs	Powerhouse	Degrees, Minutes, Seconds (Latitude)	40 ° 42' 05"	40 ° 42' 46"	Degrees, Minutes, Seconds (Longitude)	37 ° 4' 12"	37 ° 1' 44"	Decimal (Latitude)	40.701389	40.712778	Decimal (Longitude)	37.07	37.028889
Coordinate System	Diversion Weirs	Powerhouse														
Degrees, Minutes, Seconds (Latitude)	40 ° 42' 05"	40 ° 42' 46"														
Degrees, Minutes, Seconds (Longitude)	37 ° 4' 12"	37 ° 1' 44"														
Decimal (Latitude)	40.701389	40.712778														
Decimal (Longitude)	37.07	37.028889														
Conditions prior to project initiation	The project activity is a large-scale hydroelectric power plant supplying generated green electricity to the regional grid. In the absence of the project activity the same amount of electricity															

	would have been generated by a mix power plants connected to the Turkish grid which is dominated by fossil fuel-based power plants. It was found to be accordance with the applied methodology ACM0002: Grid-connected electricity generation from renewable sources v22.0/04/.
Project compliance with applicable laws, statutes and other regulatory frameworks	The validation team has verified that there are no laws or regulations in Türkiye that restrict the use of Renewable Energy to generate electricity. The project is a voluntary effort by the project proponent.
Double counting and participation under other GHG programs	The Project is not claiming from any emission trading programs or has binding emission limits. Additionally, Türkiye does not have any such programme/scheme/ policy. Declaration that the project activity will not seek credits under any other program and confirming that PP will own the Verified emission reductions, submitted by the Project Proponent has been checked/12/ to verify the same.
No double claiming with emissions trading programs or binding emission limits	The Project is not claiming from any emission trading programs or has binding emission limits. Additionally, Türkiye does not have any such programme/scheme/ policy. Declaration that the project activity will not seek credits under any other program and confirming that PP will own the Verified emission reductions, submitted by the Project Proponent has been checked/12/ to verify the same.
No double claiming with other forms of environmental credit	The Project is not claiming any other form of environmental credits. Additionally, Türkiye does not have any such programme/scheme/ policy of any other form of environmental credits. Declaration that the project activity will not seek credits under any other program and confirming that PP will own the Verified emission reductions, submitted by the Project Proponent has been checked/12/ to verify the same.
Supply chain (Scope 3) emissions double claiming	The project activity is the implementation and operation of the hydro power plant. The project doesn't affect the footprint of any products that are part of a supply chain as specified in the VCS PD/02/ section 1.17.3.
Sustainable development contributions	The validation team has reviewed the sustainable development criteria set out in the VCS PD/02/. The project activity will contribute to SDG 7, SDG 8, and SDG 13 as specified in the VCS PD/02/.

	The project activity estimates to generate 72,276 MWh/year through the implementation of the plant contributing to SDG 7, which has been verified through the generation license/27/ and contribute to an average of 33,013 tCO₂e per year emission reduction claiming SDG 13, which has been verified through the ex-ante ER calculation sheet/03/. The project activity aims to provide 27 employment due to the operation, contributing to SDG 8, which has been verified through the Social security records provided by the PP/21/.
Additional information relevant to the project	As per the applied methodology, no leakage emissions have been considered for the project activity. PP has provided two commercially sensitive documents namely “The agreement between the carbon consultant and the project owner” and “Expert reports prepared by 3rd parties”/16/, /25/ to the VVB for validation. Further, there is no additional information applicable for the project activity that may have a bearing on the eligibility of the project, the reductions or removals, or the quantification of the project’s reductions or removals.

3.2 Safeguards and Stakeholder Engagement

3.2.1 Stakeholder Engagement and Consultation

3.2.1.1 Stakeholder Identification

Item	Evidence gathering activities, evidence checked, and assessment conclusion
Stakeholder identification	The local stakeholders identified mostly include villagers living in the vicinity of the project activity. The validation team reviewed the public participation (chapter IX) from the EIA report/25/ provided by the PP, which states the nearby local people as stakeholders of the project activity. Local stakeholders are those who are directly affected by the project activities, impacting the local environment, social life, and economy. These stakeholders can be located within the boundaries of the village, district, state, or nation. Public participation was done as part

	of the EIA process to gather feedback from all relevant stakeholders, providing detailed information about the project and its potential impact. The stakeholders identified by the PP were verified from the EIA report /25/ and interviews of local stakeholders during the on-site audit /13/.
Legal or customary tenure/access rights	It has been verified through generation license/27/ that RHG Enertürk Enerji Üretim ve Ticaret A.Ş is the legal owner of the project activity who is responsible entity for development, commissioning and operation of the Umut-II 24.450 MW hydropower plant. The validation team has interviewed the local stakeholders/13/ in an unbiased fashion and confirms that there is minimal private land, no involuntary relocation or no conflicts with any stakeholders. The PP will follow national laws if resettlement of rural, urban or forest households are necessary, which includes proper land ownership and compensation as verified from the EIA Report /25/.
Stakeholder diversity and changes over time	The project activity is the implementation and operation of the hydro power plant. The stakeholders identified are mostly nearby villagers. Due to the project type, no change in social, cultural or economic diversity of stakeholders is envisaged over time.
Expected changes in well-being	The project activity is the implementation and operation of the hydro power plant, which provides renewable energy to the grid of Türkiye, hence, reducing the GHG emission to the atmosphere, enhancing energy security and reduced reliance on fossil fuels. This has been reviewed through the on-site audit/13/. PP has provided the employment records/21/, which verified the employment generated during the construction and operation of the plant. Furthermore, the PP assures that the project would not have any negative impacts on stakeholders' well-being.
Location of stakeholders	The project activity is located in the Black Sea region of Türkiye, near Akkus town, Ordu Province, Karakuş River. The Powerhouse is located close to the Gokcebayir village, and the diversion weirs are located close to the Karakus village. The local stakeholders identified mostly include villagers living in the vicinity of the project activity. The project activity is in a remote location with no direct impact on the stakeholders.

Location of resources	The project activity is the implementation and operation of a run of the river hydro power plant. It has been confirmed through on-site audit/13/ and feasibility report/16/ that water diverted from the river has been discharged to the same stream, and there is no impact habitation and property of the local stakeholders. The project activity is in a remote location with no impact on the stakeholders. In addition, there are no territories or resources owned by or to which stakeholders have customary access.
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3.2.1.2 Stakeholder Consultation and Ongoing Communication

Item	Evidence gathering activities, evidence checked, and assessment conclusion
Stakeholder engagement process	The public participation meeting consisting of nearby villagers was done as part of the EIA process on 10-April-2009 at 10.00 AM in Calca Village Niksar, Tokat and 14.00 PM in Gokcebayir Village Akkus, Ordu which was verified from the EIA report /25/. An announcement containing the place and time of the meeting was published and advertised in a local newspaper published in Ordu and Tokat provinces (published in the center and districts) and in a national newspaper for the announcement of the Public Participation Meeting to the public (EIA REPORT /25/). The date and mode of invitation were confirmed during interviews with the local stakeholders conducted by the validation team. During the meeting the attendees were informed about the project activity and its impacts through a Power Point presentation.
Consultation outcome	The public participation meeting for the Umut-II project was held at the project site, involving local stakeholders like villagers. The meeting aimed to explain the VCS project activity, gather feedback, and address any concerns. No positive or negative feedback were received from the stakeholders regarding the project. The attendees of the public participation meeting expressed that they were satisfied with the implementation of the project and have therefore no concerns either to the Hydro power plant or to the Owner of the plant, which were verified from the EIA report /25/.

Ongoing communication	Stakeholders present at the meeting were informed that any grievances or feedback regarding the project, either during construction or operation, could be directed to the person in charge. PP has provided a continuous input Grievance Expression Process Book /14/ to the Chief of Umut-II HEPP to record the grievances received from the stakeholders related to the operation of the project. PP assures that the concerns raised will be resolved with a reasonable time frame. No adverse feedback or concerns necessitating changes to the project design or monitoring plan were raised during the public participation meeting. The grievance book/14/ has been checked by the VVB and it is confirmed that no negative feedback has been received during the previous crediting period (21/02/2014 – 20/02/2024). Also, During the on-site audit, local stakeholders were interviewed and confirmed that they know about the grievance process and there is no negative feedback regarding the project.
Stakeholder input	No positive or negative inputs were received during the public participation meeting.

3.2.1.3 Free, Prior, and Informed Consent

Item	Evidence gathering activities, evidence checked, and assessment conclusion
Obtaining consent	The project activity “Umut-II Hydropower Plant, Turkey” is operation of 24.450 MW hydroelectric project owned by RHG Enertürk Enerji Üretim ve Ticaret A.Ş. PP organized a public participation meeting as part of the EIA process to obtain consent from the concerned local stakeholders. Information regarding the operation of the project activity and its potential impacts were addressed to the local stakeholders, verified from the EIA report /25/. PP did not identify any IPs, LCs or customary rights holders as stakeholders during the meeting. In addition to this, no ongoing or unresolved conflicts between the PP and the stakeholders were found. During the on-site audit/13/, local stakeholders were interviewed and there are no complaints from the locals regarding the operation of the hydro power plant.

Outcome of FPIC discussion	Since the Umut-II HEPP is situated in a remote location with no human settlement in the vicinity of the project, no impact is observed. PP did not identify any IPs, LCs or customary rights holders as stakeholders during the meeting. In addition to these, no ongoing or unresolved conflicts between the PP and the stakeholders were found. PP assures that the project activity has not encroached on land, relocated people without consent, and forced physical or economic displacement. During the on-site audit/13/, local stakeholders were interviewed and there are no complaints from the locals regarding the operation of the UMUT-II HEPP. No positive or negative feedback were received from the stakeholders regarding the project. The attendees of the public participation meeting expressed that they were satisfied with the implementation of the project and have therefore no concerns either to the Hydro power plant or to the Owner of the plant, which were verified from the EIA report /25/.
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3.2.1.4 Grievance Redress Procedure

Item	Evidence gathering activities, evidence checked, and assessment conclusion
Development process	PP has discussed the grievance procedure during the public participation meeting conducted and the local stakeholders are aware of the grievance procedure, which has been confirmed through the EIA Report /25/, interviews with the PP's representatives/13/ and interviews with the local stakeholders during the on-site audit. PP has provided a continuous input/ Grievance Expression Process Book to the Chief of Umut-II HEPP to record the grievances received from the stakeholders related to the operation of the project activity. Or the grievances can be conveyed directly to the project manager or via mukhtar.
Grievance redress procedure	The grievances can be conveyed directly to the project manager or via mukhtar. This has been confirmed through interviews with the local stakeholders. The grievance book placed at the site has been checked during the on-site audit/13/ and is found to be appropriate. PP ensures that the concerns raised will be resolved within a reasonable time frame. The procedure followed to address the concerns and grievance has been reviewed and is found to be appropriate.

3.2.1.5 Public Comments

In accordance with the requirement in clause 3.18.9 of the VCS standard, Version 4.7/06/ “All VCS projects are subject to a 30-day public comment period. The date on which the project is listed on the project pipeline marks the beginning of the project’s 30-day public comment period”.

The scope of the project activity is renewal of crediting period (RCP). Therefore, no public comment period was conducted.

Comments received	Actions taken by the project proponent	Evidence gathering activities, evidence checked, and assessment conclusion
NA	NA	NA

3.2.2 Risks to Local Stakeholders and the Environment

3.2.2.1 Management Experience

The project activity, developed and operated by RHG Enertürk Enerji Üretim ve Ticaret A.Ş., has a proven track record in managing similar projects in the host country. The management team has significant experience in executing hydro power plant projects, with a proven track record in renewable energy infrastructure, which has been verified through the project proponent’s website/41/. Some similar hydropower projects delivered by RHG Enertürk Enerji Üretim ve Ticaret A.Ş. are Erenköy HEPP, Yayla HEPP, Sümer HEPP, Tuna HEPP and Yaprak HEPP, which has been verified from the RHG Enertürk Enerji Üretim ve Ticaret A.Ş,webpage /41/.

3.2.2.2 Risk Assessment

Item	Evidence gathering activities, evidence checked, and assessment conclusion
Natural and human-induced risks to stakeholders’ wellbeing	There are no major risks identified for the project activity. The project activity involves the operation of grid connected hydro power plant as confirmed through on-site audit /13/ and feasibility study report/16/. The Hydro project is designed to minimize impact on stakeholders' rights, with thorough planning and adherence to regulatory standards including Environmental Impact Assessment Regulation No. 26939 /39/ to reduce potential negative effects on residents, the environment, and community well-being, which has been

	verified from the EIA report /25/ and EIA positive certificate /26/.
Risks to stakeholder participation	There are no major risks identified for the project activity. The project activity involves the operation of grid connected hydro power plant as confirmed through remote audit /13/ and feasibility study report/16/. Prior to the public participation meeting, stakeholders received concise information regarding the project and its agenda /25/. There were no unresolved concerns or conflicts with any relevant stakeholders. Consequently, no potential risks have been identified could impact stakeholder participation in the project. In addition, the project activity and PP adheres to Environmental Impact Assessment Regulation No. 26939 in Türkiye /39/.
Working conditions	There are no major risks identified from the project activity regarding the working conditions. During the on-site audit /13/, the working environment of the project site was observed, and the workers were provided with appropriate safety equipment. In addition, the employees of the project are trained constantly and follow health and safety rules/20/. In addition, the project activity and PP adheres to Occupational Health and Safety Regulation in Türkiye /39/.
Safety of women and girls	There are no major risks identified from the project activity regarding the safety of women and girls. The grievance register/14/ has been reviewed and no complaints has been received related to the safety of women. Also, the location of Umut-II HEPP is quite far from the nearby human settlements and therefore no risk is identified. As observed from the on-site audit /13/, currently, there are no women employees working in the UMUT-II HEPP mainly because of the remote location of the plant. In addition, Türkiye has ratified ILO conventions 100, 111, 122 and 142.
Safety of minority and marginalized groups, including children	There are no major risks identified from the project activity regarding the safety of minority and marginalized groups, including children. The grievance register/14/ has been reviewed and no complaints have been received related to the safety of minority and marginalized groups, including children, also observed from the on-site audit /13/. Also, the

	location of Umut-II HEPP is quite far from the nearby human settlements and therefore no risk is identified. Türkiye has ratified several ILO conventions promoting gender equality, labor rights, and the elimination of forced and child labor (100, 111, 122, 142). The project aligns with national gender strategies and ensures non-discrimination, social security, and formal employment agreements for all workers as confirmed during on-site audit/13/. Any complaints will be promptly addressed throughout the project's duration, and the project proponent is committed to upholding human rights and labor standards.
Pollutants (air, noise, discharges to water, generation of waste, release of hazardous materials, and chemical pesticides and fertilizers)	There are no major risks identified from the project activity regarding the pollution of the environment due to the operation of the project activity. PP will follow the Environmental Impact Assessment Regulation No. 26939/39/ in case any waste including hazardous or chemical are generated. PP follows 'Regulation on Control of Waste Oils, No. 26952/39/ and disposal of waste oil will be done by third-party facilities, as verified from the EIA report /25/, waste records /29/ and during the on-site audit /13/. Transportation of waste to disposal facilities will be carried out by licensed transporters. There are no other risks identified from the project activity regarding the pollution of the environment due to the operation of the project activity. All wastes produced from the project activity is disposed as per Turkish laws and regulations /39/. In addition, the project activity is a renewable project and has received EIA positive certificate /26/.

3.2.3 Respect for Human Rights and Equity

3.2.3.1 Labor and Work

Item	Evidence gathering activities, evidence checked, and assessment conclusion
Discrimination	Since the onset of the project activity, no employees have reported any form of discrimination. From the interviews with site personnel during the onsite audit/13/, Validation team confirms that there are no complaints on discrimination or any form of harassment from the stakeholders. Furthermore, Türkiye has ratified ILO convention 111 (Discrimination (Employment and

	Occupation) Convention, 1958), and PP ensures that the workers are socially secured by the PP and protected by employment agreements and no such activity is going to be happen in the future.
Sexual harassment	Since the onset of the project activity, no project stakeholders have reported any form of sexual harassment. From the interviews with stakeholder during the on-site audit /13/, Validation team confirms that there are no complaints on sexual or any form of harassment from the stakeholders. Furthermore, PP adheres to Penal Code of Türkiye which ensures that the workers are protected from sexual harassment and no such activity is going to be happen in the future. Also, Türkiye is a party of IPEC since 1932 and ratified ILO convention 29, 111, 138 and 182./43/.
Equal pay for equal work	Since the onset of the project activity, no employees have reported any complaints regarding unequal pay as verified from the grievance register /14/. Furthermore, PP adheres to the Turkish Labor Law-4857 /39/ which ensures that the workers are paid equally irrespective of gender and protected by employment agreements and no such activity is going to be happen in the future. In addition, Türkiye has ratified ILO 100 (The Equal Remuneration Convention), ILO 111 (The Discrimination Convention), ILO 122 (Employment Policy Convention) and ILO 142 (Human Resources Development Convention).
Gender equity in labor and work	Since the onset of the project activity, no employees have reported any complaints regarding inequity in labor and work as verified from the grievance register /14/. Furthermore, PP adheres to the Turkish Labor Law-4857 /39/ which ensures that the workers are ensured equal opportunities irrespective of gender and protected by employment agreements and no such activity is going to be happen in the future. The project aligns with the national gender strategy and ensures that no form of gender-based discrimination is practiced or tolerated, as confirmed during interviews during on-site audit. In addition, Türkiye has ratified ILO 100 (The Equal Remuneration Convention), ILO 111 (The Discrimination Convention), ILO 122 (Employment Policy Convention) and ILO 142 (Human Resources Development Convention).
Forced labor	The project activity does not involve or promote forced labor as it adheres to all laws and regulations especially Turkish Labor Law-4857 and human rights law /39/. The project activity does not involve any forced labor. No such activities were found from the interviews with the PP and local stakeholders during onsite audit/13/ or through the direct observation by the validation team. All workers are provided with social security by the project proponent and are protected under formal employment agreements. Any complaints received by the PP will be promptly addressed throughout the entire duration of the project. In addition, Türkiye has ratified ILO 100 (The Equal Remuneration Convention), ILO 111 (The

	Discrimination Convention), ILO 122 (Employment Policy Convention) and ILO 142 (Human Resources Development Convention).
Child labor	The project activity does not involve or promote child labor as it adheres to all laws and regulations especially Turkish Labor Law-4857 and human rights law /39/. The project activity does not involve any child labor. No such activities were found from the interviews with the PP and local stakeholders during onsite audit/13/ or through the direct observation by the validation team. All workers are provided with social security by the project proponent and are protected under formal employment agreements. Any complaints received by the PP will be promptly addressed throughout the entire duration of the project. In addition, Türkiye has been a party to International Programme on the Elimination of Child Labour (IPEC) by the ILO and has ratified ILO 111 (The Discrimination Convention), ILO 122 (Employment Policy Convention) and ILO 142 (Human Resources Development Convention).
Human trafficking	The project activity does not involve or promote human trafficking. No such activities were found from interviews with the PP and local stakeholders during on-site audit /13/. Furthermore, adheres to all laws and regulations especially Turkish Labor Law-4857 and human rights law /39/.

3.2.3.2 Human Rights

Risks identified	Evidence gathering activities, evidence checked, and assessment conclusion
No risk identified	The project activity is the implementation and operation of a grid connected hydro power plant and is in compliance with the international human rights, the United Nations declaration on the rights of indigenous people and ILO conventions 100 (The Equal Remuneration Convention), ILO 111 (The Discrimination Convention), ILO 122 (Employment Policy Convention) and ILO 142 (Human Resources Development Convention). Further, the PP recognizes, respects, and promotes the protection of the rights of employees and stakeholders and ensures quick action if any complaints are received. Also, Türkiye is a party of IPEC since 1932 and ratified ILO convention 29, 138 and 182./43/

3.2.3.3 Indigenous Peoples and Cultural Heritage

Risks identified	Evidence gathering activities, evidence checked, and assessment conclusion
No risk identified	The project activity is the implementation and operation of a grid connected hydro power plant. The project has been implemented on the land owned by RHG Enertürk Enerji Üretim ve Ticaret A.Ş., and there is no dispute with the stakeholders regarding the land, which has been conformed through the on-site interviews conducted by the validation team. In addition to these, the project is in a remote location and there are no cultural heritage sites in the vicinity.

3.2.3.4 Property Rights

Risks identified	Evidence gathering activities, evidence checked, and assessment conclusion
No risk identified	The project activity is the implementation and operation of a grid connected hydro power plant. The project has been implemented on the land owned by RHG Enertürk Enerji Üretim ve Ticaret A.Ş., and there are no dispute with the stakeholders regarding the land, which has been confirmed through the on-site interviews/13/ conducted by the validation team. Furthermore, no expropriation was required during the implementation of the project activity. Since the project has been operational for more than 10 years and no grievances reported in the grievance book /14/, no risk is identified in the future.

3.2.3.5 Benefit Sharing

Item	Evidence gathering activities, evidence checked, and assessment conclusion
Process used to design the	The project activity is the implementation and operation of a grid connected hydro power plant. The project has been implemented on the land owned by

benefit sharing plan	RHG Enertürk Enerji Üretim ve Ticaret A.Ş. Hence, there is no benefit sharing in the project activity. The section is not applicable.
Summary of the benefit sharing plan	Not Applicable
Approval and dissemination of benefit sharing plan	Not Applicable

3.2.4 Ecosystem Health

Item	Evidence gathering activities, evidence checked, and assessment conclusion
Impacts on biodiversity and ecosystems	No negative impact identified on the local biodiversity and ecosystem. The validation team has observed from the on-site audit/13/ and have also confirmed the same through the interviews with the site personnels. Further, the project received an EIA Positive Certificate on 17/12/2009 from the Ministry of Environment and Forestry General Directorate of Environmental Impact Assessment and Planning, Türkiye /26/. Therefore, no risk is identified in the future.
Soil degradation and soil erosion	No risk has been identified. The validation team confirmed from the on-site audit/13/ that the PP has undertaken measures to prevent soil erosion. PP monitors soil permeability, water pressure formation and seepage from time to time. Further, the project received an EIA Positive Certificate on 17/12/2009 from the Ministry of Environment and Forestry General Directorate of Environmental Impact Assessment and Planning, Türkiye /26/.
Water consumption and stress	It has been confirmed through on-site audit/13/ and EIA Report/25/ that minimum discharge has been maintained in the river stream. The plant uses the water left from the channel to generate electricity, before conveying water to the riverbed, therefore, it has no impact on water availability and does not create water stress. The project partially diverts the flow of the Karakuş River to a dam for regulation and conveyance to the

	powerhouse. The water is not consumed in the process and is subsequently released back into the channel. Also, the project plays a role in conveying water to the riverbed as a part of the irrigation channel. Hence, no risk has been identified.
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3.2.4.1 Rare, Threatened, and Endangered species

Risk identified	Evidence gathering activities, evidence checked, and assessment conclusion
Species and habitat	No negative impact identified on species and habitat. The project activity involves generation of electricity from operation of hydro power plant. The validation team confirmed from the EIA Report/25/ and EIA positive certificate /26/ that the possibility of threat to the rare and endangered species is of minimal probability.
Areas needed for habitat connectivity	Not Applicable

3.2.4.2 Introduction of Species

The project activity involves the implantation and operation of grid connected hydro power plant and does not involve introduction of any species. Hence, the section is not applicable.

Species introduced	Evidence gathering activities, evidence checked, and assessment conclusion
N/A	N/A

Existing invasive species	Evidence gathering activities, evidence checked, and assessment conclusion
N/A	N/A

Evidence gathering activities, evidence checked, and assessment conclusion
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Invasive species

N/A

3.2.4.3 Ecosystem conversion

The project activity involves the implantation and operation of grid connected hydro power plant and does not involve ecosystem conversion. Hence, the section is not applicable.

Risks Identified	Evidence gathering activities and evidence checked
Ecosystem conversion	

3.3 Application of Methodology

3.3.1 Title and Reference

As described in the VCS PD section 3.1, ACM0002: Grid-connected electricity generation from renewable sources /04/, has been applied to the Project Activity. The project uses an approved baseline and monitoring methodology, and the relevant tools prescribed therein. The validity of the versions of the applied methodology and tools were checked from CDM website.

The project has also referred to following CDM Tools:

- TOOL07 “Tool to calculate the emission factor for an electricity system” (Version 07.0)/22/
- TOOL11 “Assessment of the validity of the original/current baseline and update of the baseline at the renewal of the crediting period” (Version 3.0.1) /33/
- TOOL01 “Tool for the demonstration and assessment of additionality” (Version 07.0.0) /17/

3.3.2 Applicability

All requirements for applicability of the applied methodology i.e., ACM0002 version 22.0/04/ are satisfied. Hence, the methodology is considered completely applicable for the new crediting period, and no request for a deviation in terms of the methodology's applicability is necessary. In the VCS PD/02/, all applicable conditions are fully and accurately included. The table below presents the assessments of the applicability conditions:

Methodology ID	Applicability condition	Assessment and conclusion
ACM0002	This methodology is applicable to grid-connected renewable energy power generation project activities that: a) Install a Greenfield power plant;	The project activity “Umut-II Hydropower Plant, Turkey” is a renewable energy plant with 24.450 MW of total installed capacity, with its own reservoir, intake, monitoring,

	b) Involve a capacity addition to (an) existing plant(s); c) Involve a retrofit of (an) existing operating plants/units; d) Involve a rehabilitation of (an) existing plant(s)/unit(s); or e) Involve a replacement of (an) existing plant(s)/unit(s). f) Install a Greenfield power plant together with a grid-connected Greenfield pumped storage power plant. The greenfield power plant may be directly connected to the PSP or connected to the PSP through the grid.	and Power Purchase Agreement to supply electricity to the national grid. This has been verified through the system usage agreement/32/, Generation license/27/ and provisional acceptance certificate/09/. Therefore, the project activity is a greenfield project. Hence, the criterion is met appropriately.
ACM0002	In case the project activity involves the integration of a BESS, the methodology is applicable to grid-connected renewable energy power generation project activities that: a) Integrate BESS with a Greenfield power plant; b) Integrate a BESS together with implementing a capacity addition to (an) existing solar photovoltaic¹ or wind power plant(s)/unit(s);	The project activity is a greenfield hydro power plant and does not involve the integration of BESS. This has been verified from remote-site audit /13/, Generation license/27/, feasibility report/16/.

	c) Integrate a BESS to (an) existing solar photovoltaic or wind power plant(s)/unit(s) without implementing any other changes to the existing plant(s); d) Integrate a BESS together with implementing a retrofit of (an) existing solar photovoltaic or wind power plant(s)/unit(s). e) Integrate a BESS together with a Greenfield power plant that is operating in coordination with a PSP. The BESS is located at site of the greenfield renewable power plant.	
ACM0002	The methodology is applicable under the following conditions: a) Hydro power plant/unit with or without reservoir, wind power plant/unit, geothermal power plant/unit, solar power plant/unit, wave power plant/unit or tidal power plant/unit; b) In the case of capacity additions,	The project activity is a greenfield hydro power plant with a reservoir. The project activity does not involve any capacity additions, retrofits or replacements and does not involve the integration of BESS or PSP. This information was verified during the on-site audit /13/, provisional acceptance reports /09/, final feasibility study /16/ and generation license /27/.

	retrofits, rehabilitations or replacements (except for wind, solar, wave or tidal power capacity addition projects) the existing plant/unit must have started commercial operation prior to the start of a minimum historical reference period of five years. The reference period is used for the calculation of baseline emissions and defined in the baseline emission section. Furthermore, no capacity expansion, retrofit, or rehabilitation of the plant/unit has been undertaken between the start of this minimum historical reference period and the implementation of the project activity; c) In case of Greenfield project activities applicable under paragraph 7(a) above, the project participants shall demonstrate that the BESS was an integral part of the design of the renewable energy	
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	project activity (e.g., by referring to feasibility studies or investment decision documents); d) The BESS should be charged with electricity generated from the associated renewable energy power plant(s). Only during exigencies 2 may the BESS be charged with electricity from the grid or a fossil fuel electricity generator. In such cases, the corresponding GHG emissions shall be accounted for as project emissions following the requirements under section 5.4.4 below. The charging using the grid or using fossil fuel electricity generator should not amount to more than 2 per cent of the electricity generated by the project renewable energy plant during a monitoring period. During the time periods (e.g., week(s), months(s)) when the BESS consumes more than	
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	2 per cent of the electricity for charging, the project participant shall not be entitled to issuance of the certified emission reductions for the concerned periods of the monitoring period. e) e. In case the project activity involves PSP, the PSP shall utilize the electricity generated from the renewable energy power plant(s) that is operating in coordination with the PSP during pumping mode	
ACM0002	In case of hydro power plants, one of the following conditions shall apply: a) The project activity is implemented in existing single or multiple reservoirs, with no change in the volume of any of the reservoirs; or b) The project activity is implemented in existing single or multiple reservoirs, where the volume of the reservoir(s) is increased and the power density calculated using equation (3), is	The project activity is a green field hydro power plant resulting in a new reservoir. The power density for a hydro power project in accordance with ACM0002 methodology version 22.0/04/: $PD = \frac{Cap_{PJ} - Cap_{BL}}{A_{PJ} - A_{BL}}$ where PD: Power density of the project activity (W/m²) CAP_{PJ}: Installed capacity of the hydro power plant after the implementation of the project activity (W)

	greater than 4 W/m ² ; or	CAP _{BL} : Installed capacity of the hydro power plant before the implementation of the project activity (W). For new hydro power plants, this value is zero
	c) The project activity results in new single or multiple reservoirs and the power density, calculated using equation (3), is greater than 4 W/m ² .	AP _J : Area of the single or multiple reservoirs measured in the surface of the water, after the implementation of the project activity, when the reservoir is full (m ²)
	d) The project activity is an integrated hydro power project involving multiple reservoirs, where the power density for any of the reservoirs, calculated using equation (3), is lower than or equal to 4 W/m ² , all of the following conditions shall apply:	AB _L : Area of the single or multiple reservoirs measured in the surface of the water, before the implementation of the project activity, when the reservoir is full (m ²). For new reservoirs, this value is zero. For the VCS Project activity
	I. The power density calculated using the total installed capacity of the integrated project, as per equation (4), is greater than 4 W/m ² ;	“Umut-II Hydropower plant, Turkey”, which had created a new reservoir, total installed capacity is 24.450 MW /09/, /27/and the area of reservoir is 87,327 m ² /24/, /30/.
	II. Water flow between reservoirs is not used by any other hydropower unit which is not a part of the project activity;	Therefore, $PD = (24.450 \times 10^6 W - 0) / (87,327 - 0) = 280 W/m^2$
	III. Installed capacity of the power plant(s) with power density lower than or equal to 4 W/m ² shall be;	So, Power density is 280 W/m ² , and it is greater than 4 W/m ² . The project activity complies with bullet (c) of Para 09 of ACM0002 methodology/04/. The applicability condition is found to be met.
	a) Lower than or equal to 15 MW; and	
	b) Less than 10 per cent of the total installed	

	capacity of integrated hydro power project.	
ACM0002	In the case of integrated hydro power projects, project proponent shall: a) Demonstrate that water flow from upstream power plants/units spill directly to the downstream reservoir and that collectively constitute to the generation capacity of the integrated hydro power project; or b) b. Provide an analysis of the water balance covering the water fed to power units, with all possible combinations of reservoirs and without the construction of reservoirs. The purpose of water balance is to demonstrate the requirement of specific combination of reservoirs constructed under CDM project activity for the optimization of power output. This	The project is not an integrated hydro power project as verified from the generation license /27/ and feasibility study /16/. Therefore, the condition is not applicable.

	demonstration has to be carried out in the specific scenario of water availability in different seasons to optimize the water flow at the inlet of power units. Therefore, this water balance will take into account seasonal flows from river, tributaries (if any), and rainfall for minimum of five years prior to the implementation of the CDM project activity.	
ACM0002	In the case of PSP, the project participants shall demonstrate in the PDD that the project is not using water which would have been used to generate electricity in the baseline.	The project activity does not involve PSP as verified from the on-site audit /13/, feasibility study /16/.
	The methodology is not applicable to: a) Project activities that involve switching from fossil fuels to renewable energy sources at the site of the project activity, since in this case the baseline may be the continued use of fossil fuels at the site;	This condition is not applicable to the project activity as it does not involve: a) Switching from fossil fuels to renewable energy sources at the site of the project activity. b) Biomass fired plants. Hence this criterion is not applicable.

	b) b. Biomass fired power plants/units.	
ACM0002	In the case of retrofits, rehabilitations, replacements, or capacity additions, this methodology is only applicable if the most plausible baseline scenario, as a result of the identification of baseline scenario, is “the continuation of the current situation, that is to use the power generation equipment that was already in use prior to the implementation of the project activity and undertaking business as usual maintenance”.	The project is not a retrofit, rehabilitations, replacements or capacity addition; hence this applicability criterion is not relevant.
CDM Tool 07	This tool may be applied to estimate the OM, BM and/or CM when calculating baseline emissions for a project activity that substitutes grid electricity that is where a project activity supplies electricity to a grid or a project activity that results in savings of electricity that would have been provided by the grid (e.g. demand-side energy efficiency projects).	This project involves generating electricity from large-scale hydro power plant where generated electricity is delivered to the grid and hereby displaces emissions which might have generated from the production of equivalent amount of power from fossil fuel thermal power plants which exists in the baseline scenario. This has been verified through the system usage agreement/32/, and generation license/27/. EF values have been adopted from “Türkiye National Network Emission Factor Data Sheet” which considers only grid connected plants

		/34/. Therefore, the applicability criteria is found to be met by evaluating the amount of baseline emission by considering OM, BM and CM emission factors.
CDM Tool 07	Under this tool, the emission factor for the project electricity system can be calculated either for grid power plants only or, as an option, can include off-grid power plants. In the latter case, two sub-options under the step 2 of the tool are available to the project participants, i.e., option IIa and option IIb. If option IIa is chosen, the conditions specified in “Appendix 2: Procedures related to off-grid power generation” should be met. Namely, the total capacity of off-grid power plants (in MW) should be at least 10 per cent of the total capacity of grid power plants in the electricity system; or the total electricity generation by off-grid power plants (in MWh) should be at least 10 per cent of the total electricity generation by grid power plants in the electricity system; and that factors which negatively affect the reliability and stability of the grid are primarily due to constraints in generation and not to other aspects such as transmission capacity.	The project activity has chosen the option to calculate the emission factor for grid power plants only. The point has been assessed in detail under section 3.3.6 of the report. Thus, criterion is found to be met.

CDM Tool 07	In case of CDM projects the tool is not applicable if the project electricity system is located partially or totally in an Annex I country.	This project involves generating electricity from large scale hydro power plant located in Türkiye which is an Annex I country as verified from UNFCCC website/10/. However, the scope is the renewal of crediting period and project is already registered under VCS. Additionally, the applicability condition is with respect to the CDM project activity and the project is under VCS program Thus, the applicability criteria are found to be not applicable.
CDM Tool 07	Under this tool, the value applied to the CO ₂ emission factor of biofuels is zero.	This project involves generating electricity from large scale hydro power plants and does not involve emissions from biofuels. The generated electricity from the project activity is delivered to the national grid of Türkiye. Thus, the applicability criteria are found to be met.
CDM Tool 11	This tool provides a stepwise procedure to assess the continued validity of the baseline and to update the baseline at the renewal of a crediting period, as required by paragraph 49 (a) of the modalities and procedures of the clean development mechanism. The tool consists of two steps. The first step provides an approach to evaluate	The scope of the project activity is renewal of crediting period. Therefore, the tool is applicable and has been applied by the PP to assess whether the current baseline is valid or invalid.

	whether the current baseline is still valid for the next crediting period. The second step provides an approach to update the baseline in case that the current baseline is not valid anymore for the next crediting period. The steps are elaborated in section 3.3.4 below of the report.	
CDM Tool 01	The use of the “Tool for the demonstration and assessment of additionality” is not mandatory for project participants when proposing new methodologies. Project participants may propose alternative methods to demonstrate additionality for consideration by the Executive Board. They may also submit revisions to approved methodologies using the additionality tool.	This project involves generating electricity from large scale hydro power plants. The scope of the project activity is renewal of crediting period. Additionality has been already demonstrated during the registration of the project activity, as verified from the registered PD /30/.
CDM Tool 01	Once the additionality tool is included in an approved methodology, its application by project participants using this methodology is mandatory.	This project involves generating electricity from large scale hydro power plants. The scope of the project activity is renewal of crediting period. Additionality has been already demonstrated during the registration of the project activity as verified from the registered PD /30/.

3.3.3 Project Boundary

As per para 25 of ACM0002 v22.0 “The spatial extent of the project boundary includes the project power plant/unit and all power plants/units connected physically to the electricity system that

the CDM project power plant is connected to.” The project boundary of the project activity includes the hydro power plant, transformer substation, electricity meters, and the national grid of Türkiye as observed from the single line diagram/19/. The electricity generated by the project activity is supplied to national grid of Türkiye. The validation team was able to confirm that all the identified emission sources which are impacted by the project activity are addressed by the approved methodology ACM0002. version 22.0/04/. Details on gases included in project boundary are given below:

Source	Gas	Included?	Justification/Explanation
Baseline	CO ₂	Yes	The major emission source. The GHG emission reductions are achieved by displacing the electricity generated by fossil fuel dominated power plants.
	CH ₄	No	Excluded for simplification. This is conservative.
	N ₂ O	No	Excluded for simplification. This emission source is assumed to be very small.
Project activity	CO ₂	No	Not Applicable, as project activity is a hydro power plant
	CH ₄	No	
	N ₂ O	No	
	CO ₂	No	Not Applicable, as project activity is a hydro power plant
	CH ₄	No	
	N ₂ O	No	
	Low GWP hydrocarbon/refrigerant	No	Not Applicable, as project activity is a hydro power plant.
	CO ₂	No	The project has diesel based 1 DG set available at the site for emergency purposes. The
	CH ₄	No	

Source	Gas	Included?	Justification/Explanation
electricity generation in solar thermal power plants and geothermal power plants	N ₂ O	No	consumption has been checked and is found to be very less, hence, emissions are negligible.
For hydro power plants, emissions of CH ₄ from the reservoir	CO ₂	No	The Project does not generate any form of CO ₂ . In urgent cases, a diesel-powered generator will be operated for consumption of the personnel and the building, which is negligible.
	CH ₄	No	The power density of the project activity is greater than 10 W/m ² , hence emissions from water reservoirs are equal to “0” as per paragraph 47 of the applied methodology, ACM0002, V22.0. Detailed assessment is provided in section 3.3.6 of this report
	N ₂ O	No	Minor emission source.
Charging of BESS using electricity from the grid or from fossil fuel electricity generators.	CO ₂	No	Not Applicable, as project activity is a hydro power plant
	CH ₄	No	
	N ₂ O	No	
Utilization of electricity from grid or from fossil fuel generators by PSP for pumped mode.	CO ₂	No	Not Applicable, as project activity is a hydro power plant
	CH ₄	No	
	N ₂ O	No	
For PSP, emissions of CH ₄ from the reservoir	CO ₂	No	The project activity does not involve pumped storage.
	CH ₄	No	
	N ₂ O	No	

3.3.4 Baseline Scenario

As per the para 27 of the applied methodology ACM0002 Version 22.0/04/ the baseline scenario is defined as “If the project activity is the installation of a Greenfield power plant with or without a BESS as described under paragraph 5(a) or paragraph 7(a) or paragraph 7(e) above, the

baseline scenario is electricity delivered to the grid by the project activity that would have otherwise been generated by the operation of grid-connected power plants and by the addition of new generation sources, as reflected in the combined margin (CM) calculations described in TOOL07."/22/

The project activity is a new 24.450 MW_e hydropower plant commissioned as a distinct unit and does not impact the electricity generation potential for existing projects and has a separate reservoir, intake, monitoring and system usage agreement /32/ to supply electricity to national grid of Türkiye.

The baseline scenario for the project activity is generation of same amount of electricity through operation of existing power plant connected to the national grid of Türkiye. This baseline scenario is correctly identified by the project proponent in the VCS PD/02/. The proportion of the generation through operational power plants and newly build power plants can be addressed by determining the Combined Margin of the grid in accordance with CDM Tool 7 version 07.0/22/.

Para 18 of ACM0002 v22.0/04/ referred to Tool 11: Assessment of the validity of the original/current baseline and update of the baseline at the renewal of the crediting period, Version 3.0.1/33/ to assess the current baseline of the project activity during the project renewal.

The tool stipulates the following steps to be carried out:

Step 1: Assess the validity of the current baseline for the next crediting period

Step 1.1: Assess compliance of the current baseline with relevant mandatory national and/or sectoral policies

The baseline scenario remains unchanged which is grid connected electricity generation plants and is in compliance with all the relevant mandatory national and/or Sectoral policies. This has been verified through the system usage agreement/32/, generation license/27/ and EIA positive certificate/26/.

Step 1.2: Assess the impact of circumstances

New circumstances in Türkiye impact the grid's emission factor and, thus, on the project activity's current baseline emissions. Accordingly, the operating margin and build margin emission factors for the second crediting period have been updated in line with the latest version of the "Türkiye's National Electric Grid Emission Factor" for 2022, published on 26 November 2024 /34/, which is the latest available data published by official sources.

Step 1.3: Assess whether the continuation of use of current baseline equipment(s) or an investment is the most likely scenario for the crediting period for which renewal is requested

There hasn't been any changes made to the equipment's used and neither has any investments been made from the time of validation. The expected operational lifetime of the project is 48 years, 8 months, and 4 days as verified from the generation license issued by the EMRA (Energy

Market Regulatory Authority) /27/. The same has been verified through the generation license/27/, provisional acceptance reports/09/ and interview with the site personnel.

Step 1.4: Assessment of the validity of the data and parameters

The project has two main parameters for the calculation of emission reduction: Electricity generation and Emission factor, which is found to be appropriate as per the applied methodology. The emission factor is an ex-ante parameter and is not monitored during the first crediting period.

PP has updated the emission factor of the project during the second crediting period in line with the applied methodology/04/ and Tool 07/22/. The project utilizes emission factor values obtained from the latest available “Türkiye ’s National Electric Grid Emission Factor” published by Ministry of Energy and Resources for the year 2022/34/, which utilizes Tool 07 to calculate the emission factor for the host country Türkiye and is the latest available at the time of validation and hence, is found to be appropriate.

Step 2: Update the current baseline and the data and parameters

As evident from the explanation provided above the baseline scenario remains unchanged. The operating margin and build margin emission factors for the second crediting period have been updated in line with the latest version of the “Türkiye ’s National Electric Grid Emission Factor” for 2022, published on 26 November 2024 /34/ available at the time of PD submission for renewal. The PP has calculated the combined margin emission factor ($EF_{grid,CM,y}$) from these updated emission factors, which in turn is used to calculate the baseline emissions.

The validation team confirms that:

- (a) All the assumptions and data used by the PP are listed in the VCS PD/02/, including their references and sources.
- (b) All documentation used is relevant for establishing the baseline scenario and correctly quoted and interpreted in the VCS PD/02/.
- (c) Assumptions and data used in the identification of the baseline scenario are justified appropriately, supported by evidence and can be deemed reasonable.
- (d) Relevant national and/ or sectoral policies and circumstances are considered and listed in the VCS PD/02/.
- (e) The approved baseline methodology/04/ has been correctly applied to identify the most reasonable baseline scenario and the identified baseline scenario reasonably represents what would occur in the absence of the proposed project activity.

The baseline scenario identified by the PP was found to be in accordance with applied methodology and the registered PD/30/. Therefore, PP’s decision not to make changes in the current baseline was found to be acceptable by the validation team.

3.3.5 Additionality

As per the section 3.9.8 of the VCS standard v4.7/06/, a full reassessment of additionality is not required when renewing the project crediting period unless otherwise specified in the methodology. However, regulatory surplus shall be demonstrated in accordance with the requirements set out in the VCS Program rules and the project description shall be updated accordingly.

The project activity is currently undergoing the validation of renewal of crediting period; therefore, the full reassessment of additionality is not required. However, PP has demonstrated the regulatory surplus by confirming that the implementation of the project activity is a voluntary initiative as taken by the project proponent and is not mandatory or a legal requirement.

3.3.6 Quantification of GHG Emission Reductions and Carbon Dioxide Removals

The GHG emissions reduction calculations are transparently documented and appropriate assumptions regarding the expected amount of electricity generated have been used to forecast emission reductions for the project activity. In according to the applied methodology ACM0002 v22.0/04/ the formulae applied in the VCS PD/02/ for the emission reductions (ER_y) calculation by the project activity is the difference between the baseline emissions (BE_y) and the project emissions (PE_y) which is expressed as follows:

Emission Reductions:

As per para 72 of the applied methodology,

$$ER_y = BE_y - PE_y$$

ER_y =Emission reductions in year y (tCO₂e/yr)

BE_y =Baseline emissions in year y (tCO₂/yr)

PE_y =Project emissions in year y (tCO₂e/yr)

According to para 57 of the applied methodology/04/, the Baseline emissions are calculated as:

$$BE_y = EG_{PJ,y} * EF_{grid, CM,y}$$

BE_y Baseline Emissions in year y (tCO₂).

$EG_{PJ, y}$ Quantity of net electricity generation that is produced and fed into the grid as a result of the implementation of the CDM project activity in year y (MWh/yr).

$EF_{grid,CM,y}$ Combined margin CO₂ emission factor for grid connected power generation in year y calculated using the latest version of the TOOL07 “Tool to calculate the emission factor for an electricity system” (tCO₂/MWh). The value for the combined margin is 0.4568 tCO₂e/MWh.

Project emissions:

As per para 40 of the applied methodology, “For most renewable energy project activities, $PE_y = 0$. However, for the following categories of project activities, project emissions have to be considered following the procedure described in the most recent version of “ACM0002: Grid-connected electricity generation from renewable sources”:

- (a) Emissions related to the operation of geothermal power plants (e.g. non condensable gases, electricity/fossil fuel consumption);
- (b) Emissions from water reservoirs of hydro power plants.”

Hence, as per para 40 of ACM0002/04/,

$$PE_y = PE_{FF,y} + PE_{GP,y} + PE_{HP,y} + PE_{BESS,y} + PE_{PSP,y}$$

Where,

PE_y	=	Project emissions in year y (tCO ₂ e/yr)
$PE_{FF,y}$	=	Project emissions from fossil fuel consumption in year y (tCO ₂ /yr)
$PE_{GP,y}$	=	Project emissions from the operation of dry, flash steam or binary geothermal power plants in year y (t CO ₂ e/yr)
$PE_{HP,y}$	=	Project emissions from water reservoirs of hydro power plants in year y (tCO ₂ e/yr)
$PE_{BESS,y}$	=	Project emissions from charging of a BESS using electricity from the grid or from fossil fuel electricity generators (t CO ₂ e/yr)
$PE_{PSP,y}$	=	Project emissions from utilizing electricity from the grid for pumping operation of PSP in excess to the production

of the renewable power plant operating in coordination with the PSP (t CO₂e/yr)

Since, the project activity is the implementation and operation of the greenfield hydro power plant. PE_{GP,y}, PE_{BESS,y} and PE_{PSP,y} is considered as zero, which is found to be appropriate.

It has been verified during the on-site audit /13/ that there is a diesel generator present at site, which is used only for emergency purposes. As per para 42 of the applied methodology/04/, For all renewable energy power generation project activities, emissions due to the use of fossil fuels for backup generator can be neglected. Hence, PE_{FF,y} is considered as zero, which is found to be appropriate and in line with the applied methodology/04/.

Emissions from water reservoirs of hydro power plants and pumped storage projects (PE_{HP,y}):

As per para 46 of ACM0002/04/, “The power density (PD) of the project activity is calculated as follows:”

$$PD = \frac{Cap_{PJ} - Cap_{BL}}{A_{PJ} - A_{BL}}$$

Where,

PD = Power density of the project activity (W/m²)

Cap_{PJ} = Installed capacity of the hydro power plant after the implementation of the project activity (W)

Cap_{BL} = Installed capacity of the hydro power plant before the implementation of the project activity (W). For new hydro power plants, this value is zero

A_{PJ} = Area of the reservoir measured in the surface of the water, after the implementation of the project activity, when the reservoir is full (m²)

A_{BL} = Area of the reservoir measured in the surface of the water, before the implementation of the project activity, when the reservoir is full (m²). For new reservoirs, this value is zero.

The installed capacity of the power plant is 24.450 MW, which has been verified through the generation license/27/, and provisional acceptance reports/09/. The value of Cap_{BL} and A_{BL} is considered as 0, since this is a new reservoir, which has been verified through the EIA Report/25/ and feasibility report/16/.

The value of the A_{PJ} has been taken as 87,327 m², which has been verified through the survey of the area of reservoir conducted by 3rd party/24/ and feasibility report /16/, and hence, found to be appropriate.

Therefore, the value of the power density comes out to be 280 W/m². Hence, as per para 47 of the ACM0002 v22.0/04/, PE_{HP,y} is equal to zero.

Leakage emissions:

The leakage emission for the project activity is considered zero, as there are no activities which will lead to emission, and it is in accordance with the applied methodology ACM0002 v22.0/04/.

Therefore, $LE_y = 0$

Emission reductions:

The baseline emissions for the project activity are calculated as:

$$BE_y = EG_{PJ,y} \times EF_{grid,CM,y}$$

Where,

$$EG_{PJ,y} = EG_{PJ,facility,y} = 72,276 \text{ MWh/year}$$

$$EF_{grid,CM,y} = EF_{CM,y} = 0.4568 \text{ tCO}_2\text{e/MWh}$$

Hence

$$BE_y = 72,276 \text{ MWh/yr} \times 0.4568 \text{ tCO}_2\text{/MWh}$$

$$BE_y = 33,013 \text{ tCO}_2\text{e/year}$$

$$ER_y = BE_y - PE_y$$

$$ER_y = 33,013 - 0$$

$$ER_y = 33,013 \text{ tCO}_2\text{e/year}$$

Vintage period	Estimated baseline emissions (tCO ₂ e)	Estimated project emissions (tCO ₂ e)	Estimated leakage emissions (tCO ₂ e)	Estimated reduction VCU (tCO ₂ e)	Estimated removal VCU (tCO ₂ e)	Estimated total VCUs (tCO ₂ e)
21 February 2024 – 31 December 2024	28,413	0	0	28,413	0	28,413
01 January 2025 – 31 December 2025	33,013	0	0	33,013	0	33,013
01 January 2026 – 31 December 2026	33,013	0	0	33,013	0	33,013

Vintage period	Estimated baseline emissions (tCO ₂ e)	Estimated project emissions (tCO ₂ e)	Estimated leakage emissions (tCO ₂ e)	Estimated reduction VCU's (tCO ₂ e)	Estimated removal VCU's (tCO ₂ e)	Estimated total VCUs (tCO ₂ e)
01 January 2027 – 31 December 2027	33,013	0	0	33,013	0	33,013
01 January 2028 – 31 December 2028	33,013	0	0	33,013	0	33,013
01 January 2029 – 31 December 2029	33,013	0	0	33,013	0	33,013
01 January 2030 – 31 December 2030	33,013	0	0	33,013	0	33,013
01 January 2031 – 31 December 2031	33,013	0	0	33,013	0	33,013
01 January 2032 – 31 December 2032	33,013	0	0	33,013	0	33,013
01 January 2033 – 31 December 2033	33,013	0	0	33,013	0	33,013
01 January 2034 – 20 February 2034	4,600	0	0	4,600	0	4,600
Total	330,130	0	0	330,130	0	330,130

The average annual emission reduction is 33,013 tCO₂e per year over a 10-year renewal crediting period.

The Validation team confirms:

- All assumptions and data used by the Project Proponents are listed in the PD/02/, including their references and sources;
- All documentation used by Project Proponents as the basis for assumptions and source of data is correctly quoted and interpreted in the PD;
- All values used in the PD are considered reasonable in the context of the proposed project activity;
- The baseline methodology and corresponding tool(s) have been applied correctly to calculate project emissions, baseline emissions, leakage and emission reductions;

All estimates of the baseline emissions can be replicated using the data and parameter values provided in the PD.

3.3.7 Methodology Deviations

The project activity has used approved methodology, ACM0002 version 22.0/04/ and all the applicability conditions and methodological requirements have been completely followed by the PP. The validation team has confirmed that the Project activity has not been subjected to any methodology deviations.

3.3.8 Monitoring Plan

The project applies the monitoring methodology ACM0002 version 22.0/04/ for monitoring emission reduction which is valid for submission. The project activity meets all the requirements as stated in the methodology and provided all the parameters in the revised version of the VCS PD/02/ in terms of ex- ante and ex-post in the respective section of the PD.

Data and parameters fixed Ex-ante:

- **Combined Margin CO₂ emission factor for the project electricity system in year y ($EF_{grid,CM,y}$):**

The value for Combined margin emission factor is taken as 0.4568 tCO₂e/MWh. This has been calculated using equation 16 of CDM TOOL 07 Version 7.0/22/. The value of operating margin and Build margin has been taken directly through the Turkish Emission factor datasheet, published by Ministry of Energy and Natural Resources/34/. The approach taken is in accordance with the CDM TOOL 07 and is found acceptable by the validation team.

$$EF_{grid,OM,y} = 0.7108 \text{ tCO}_2\text{e/MWh}$$

$$EF_{grid,BM,y} = 0.3721 \text{ tCO}_2\text{e/MWh}$$

As per equation 16 of CDM TOOL 07 Version 7.0/22/:

$$EF_{grid,CM,y} = EF_{grid,OM,y} \times w_{OM} + EF_{grid,BM,y} \times w_{BM}$$

$$EF_{grid,CM,y} = 0.7108 \times 0.25 + 0.3721 \times 0.75 = 0.4568 \text{ tCO}_2\text{e/MWh}$$

The validation team has verified the values fixed, values are taken from the reliable sources (all the sources are mentioned in Appendix 2 of the report and therefore it has been accepted by the validation team.

Data and parameter monitored:

To monitor the emission reduction of the project activity, the net electricity exported by the project activity to the grid is multiplied with the grid emission factor, to further evaluate the emission reduction achieved (ER_y). Therefore, the monitoring plan includes monitoring of following parameters:

- **Quantity of net electricity generation supplied by the project plant/unit to the grid in year y ($EG_{PJ, facility, y}$):**

The value of the parameter will be continuously monitored and monthly reported through the electricity meters (main and spare meters). The value of the parameter is calculated by subtracting the net export with the net import by the project activity. The electricity meters of UMUT – I, UMUT – II, and UMUT – III has been connected to the same electricity meters. Hence, PP has applied apportioning techniques based on the installed capacity of the all the power plants, The description of the apportioning has been properly described in the VCS PD/02/, which has been assessed and is found to be appropriate. The electricity supplied and withdrawn from the grid will be monitored via EPIAS records. The expected net electricity generation for the crediting period (21-February 2024 to 20- February 2034) is 722,759 MWh and expected annual net generation is 72,276 MWh as verified from the generation license /27/ and the ER sheet /03/.

Two tri-vector bi-directional energy meters of accuracy class 1S are installed at the site namely main meter and spare meter which are going to be calibrated once every ten years by the representatives of TEIAS (Turkish Electricity Transmission Corporation) in accordance with Article 9 of the MEASUREMENT AND MEASURING INSTRUMENTS INSPECTION REGULATION/35/. The specifications of the energy meters are given below which has been verified from the meter replacement certificate/28/ and during on-site audit /13/ and is found to be appropriate. The source of data for the parameter is EPIAS Records/31/, and will be cross-checked using invoice and TEIAS protocols/36/.

Properties	Main Meter	Spare Meter
Brand	EMH	EMH
Type	LZQJ-XC	LZQJ-XC
Class	1S	1S
Serial No	9420180	9420181
Test Date	23/07/2020	23/07/2020

Installation Date	14/10/2020	14/10/2020
Calibration Due	14/10/2030	14/10/2030

Therefore, all the parameters mentioned in the revised VCS PD/02/ are found to be appropriate and in line with the applied methodology/04/. Hence, accepted by the validation team.

The validation team confirms that:

- All the values used from official sources and the authenticity of sources has been verified by the validation team and confirms that all relevant parameters to calculate the GHG emissions reductions of the project have been sufficiently considered and the value of the ex-ante fixed parameter used for emission reduction calculation i.e. grid emission factor has been determined conservatively and the estimation of ex-post parameters are reasonable.
- The validation team considers that the monitoring plan has complied with the requirements in the approved methodology/04/. The monitoring plan has been found to be in compliance with the requirements of the applied methodology. The monitoring plan will give opportunity for real measurements of achieved emission reductions.

The validation team considers that monitoring arrangements described in the monitoring plan is feasible within the project design and the PP will be capable to implement the monitoring plan.

3.4 Non-Permanence Risk Analysis

It is not an AFOLU project activity, hence it is not applicable.

4 VALIDATION OPINION

4.1 Validation Summary

Earthood Services Limited (Earthood) performed the validation of the emission reductions for the VCS project activity “Umut-II Hydropower Plant, Turkey” (VCS ID-1011) for the second crediting period 21/02/2024 to 20/02/2034. RHG Enertürk Enerji Üretim ve Ticaret A.Ş. is responsible for the collection of data in accordance with the monitoring plan and the reporting of GHG emissions reductions from the project activity. It is our responsibility as VVB to express an independent validation statement on the reported GHG emission reductions from the project activity.

Earthood commenced the validation based on the baseline and monitoring methodologies ACM0002 version 22.0 /04/, the monitoring plan contained in the PD and VCS Program

Guidelines version 4.4 as per the process described under Section 2 of this report. This validation was carried out using a risk-based approach. 06 Corrective Action Requests (CAR), 02 Clarification Requests (CL) and 01 Forward Action Requests (FAR) were raised and successfully closed during the validation.

The VVB has carried out this validation with a reasonable level of assurance as stated in section 1.3 above. Further, the verification of the GHG statement was conducted in accordance with ISO 14064-3).

4.2 Validation Conclusion

Earthood has performed the validation by obtaining evidence, conducting a comprehensive on-site audit, and requesting additional information. The Project Proponent (PP) has undertaken a detailed calculation of estimated emission reductions in accordance with the applied methodology ACM0002 version 22.0. The provided explanations by the PP are deemed necessary by Earthood to give a strong level of assurance that the projected GHG emissions reductions are not only attainable but will, in fact, be achieved. This is exemplified by the commitments to high standards of accuracy and integrity shown throughout the validation process.

As a result of the validation, the validation team confirms that:

- The project fulfils criteria of VCS standard Version 4.7/06/.
- The project is in line with all relevant VCS requirement.
- The project additionality is sufficiently justified in the VCS PD/02/.
- The monitoring plan is transparent, adequate and in line with applied baseline and monitoring methodology of ACM0002 version 22.0/04/.
- The calculation of the emission reductions from the project activity is carried out in a transparent and conservative manner, so that the calculated emission reductions of 330,130 tCO_{2e} is most likely to be achieved within the 10 years of crediting period.

No restrictions or uncertainties were identified related to the validation.

Crediting period: From 21-February-2024 to 20- February -2034

Validated estimated GHG emission reductions and carbon dioxide removals for the project crediting period:

Vintage period	Estimated baseline emissions (tCO ₂ e)	Estimated project emissions (tCO ₂ e)	Estimated leakage emissions (tCO ₂ e)	Estimated reduction VCU (tCO ₂ e)	Estimated removal VCU (tCO ₂ e)	Estimated total VCUs (tCO ₂ e)
21 February 2024 – 31 December 2024	28,413	0	0	28,413	0	28,413
01 January 2025 – 31 December 2025	33,013	0	0	33,013	0	33,013
01 January 2026 – 31 December 2026	33,013	0	0	33,013	0	33,013
01 January 2027 – 31 December 2027	33,013	0	0	33,013	0	33,013
01 January 2028 – 31 December 2028	33,013	0	0	33,013	0	33,013
01 January 2029 – 31 December 2029	33,013	0	0	33,013	0	33,013
01 January 2030 – 31 December 2030	33,013	0	0	33,013	0	33,013
01 January 2031 – 31 December 2031	33,013	0	0	33,013	0	33,013
01 January 2032 – 31 December 2032	33,013	0	0	33,013	0	33,013

Vintage period	Estimated baseline emissions (tCO ₂ e)	Estimated project emissions (tCO ₂ e)	Estimated leakage emissions (tCO ₂ e)	Estimated reduction VCU's (tCO ₂ e)	Estimated removal VCU's (tCO ₂ e)	Estimated total VCU's (tCO ₂ e)
01 January 2033 – 31 December 2033	33,013	0	0	33,013	0	33,013
01 January 2034 – 20 February 2034	4,600	0	0	4,600	0	4,600
Total	330,130	0	0	330,130	0	330,130

Approved By:



Dr. Kaviraj Singh

Chief Executive Officer

Date: 21/06/2025

Earthood Services Limited

Place: Gurgaon, Haryana

APPENDIX 1: COMMERCIALLY SENSITIVE INFORMATION

There are no commercially sensitive information

Section	Information	Justification	Assessment method and conclusion
<u>NA</u>	NA	NA	NA

APPENDIX 2: DOCUMENTS REFERRED

S.No.	Title of Document	Version	Date
1.	Initial VCS PD	v1.0	
2.	Final VCS PD	V3.0	20/05/2025
3.	Estimated ER sheet	Pertaining to latest PD	-
4.	ACM0002: Grid-connected electricity generation from renewable sources	V22.0	-
5.	VCS Project ID- https://registry.verra.org/app/projectDetail/VCS/771	-	-
6.	VCS Standard	4.7	16/04/2024
7.	VCS Program Guide	4.4	29/08/2023
8.	VCS Validation and verification manual	3.2	19/10/2016
9.	Provisional Acceptance Certificate	-	21/02/2014
10.	Türkiye: Type of Party https://unfccc.int/node/61221	-	Last accessed on 18/05/2025
11.	Geographic location: google earth KML File	-	-
12.	Declaration/ confirmation by PP that the project activity will not seek credits under any other program and confirming that PP will own the Verified emission reductions	-	-
13.	On-site audit	-	27/06/2024
14.	Grievance Register		
15.	VCS Program Definitions	4.5	16/04/2024
16.	Feasibility Study Report	-	December 2019
17.	TOOL01 “Tool for the demonstration and assessment of additionality”	Version 07.0.0	
18.	https://www.un.org/development/desa/dpad/wp-content/uploads/sites/45/publication/ldc_list.pdf	-	Accessed on 18/05/2025
19.	Single Line Diagram of UMUT Hydro power plants	-	
20.	Health & Safety Training Records	-	-
21.	Employment Records, Social security records	-	-
22.	Tool 07: Tool to calculate the emission factor for an electricity system	Version 07.0	-
23.	Registration and Issuance Process	Version 4.6	16/10/2024
24.	Survey for Area of Reservoir by YOLSU International Engineering and Consulting Services Co.	-	14/08/2024
25.	Environmental Impact Assessment (EIA) Report	-	28/09/2009
26.	EIA Positive Certificate	-	12/05/2008
27.	Generation License	-	01/05/2008
28.	Meter Replacement Certificate	-	14/10/2020
29.	Waste Records	-	
30.	Registered VCS PD (Crediting Period:04/10/2012 to 03/10/2022)	V1.4	18/01/2012
31.	EPIAS Records	-	-

32.	System Usage agreement	-	24/11/2021
33.	Tool 11: Assessment of the validity of the original/current baseline and update of the baseline at the renewal of the crediting period	V 3.0.1	02/03/2012
34.	Türkiye 's National Electric Grid Emission Factor for 2022	-	26/11/2024
35.	MEASUREMENT AND MEASURING INSTRUMENTS INSPECTION REGULATION	-	-
36.	TEIAS Protocols	-	-
37.	Verra website	-	-
38.	CDM PS for PA	v3.0	
39.	Laws and regulations in Türkiye - Human Rights – Ministry of Foreign Affairs (1987) https://www.mfa.gov.tr/%C4%B0nsan-haklar%C4%B1.en.mfa - LAW ON EQUAL OPPORTUNITIES FOR WOMEN AND MEN No. 5840 (24-March-2009) - Turkish Labor Law-4857 https://turkishlaborlaw.com/turkish-labor-law-4857/4857-labor-law-english-plain-text/ - WASTE MANAGEMENT REGULATION https://www.resmigazete.gov.tr/eskiler/2015/04/20150402-2.htm - PENAL CODE OF TURKEY – Article 105 (15-February-2016) https://www.venice.coe.int/webforms/documents/default.aspx?pdffile=CDL-REF(2016)011-e - Environmental Law No. 2872 (09-August-1983) https://www.mevzuat.gov.tr/mevzuatmetin/1.5.2872.pdf - Electricity Market Law No. 4628 (03-March-2001) https://www.mevzuat.gov.tr/MevzuatMetin/1.5.4628-20080709.pdf - Law on Utilization of Renewable Energy Resources for the Purpose of Generating Electrical Energy No. 5346 (18-May-2005) https://www.mevzuat.gov.tr/mevzuatmetin/1.5.5346.pdf - Energy Efficiency Law No. 5627 (02-May-2007) https://www.mevzuat.gov.tr/mevzuatmetin/1.5.5627.pdf		
40.	Connection Agreement	-	08/08/2013

41.	RHG Enertürk Enerji Üretim ve Ticaret A.Ş.,webpage https://www.enerturk.com/kurumsal/faaliyetlerimiz/elektrik-uretim/hidroelektrik-santralleri-hes	-	-
42.	Verra projects Portal https://registry.terra.org/app/search/VCS/All%20Projects	-	-
43.	ILO Webpage – Turkey https://www.ilo.org/regions-and-countries/europe-and-central-asia/t%C3%BCrkiye-0		Accessed on 10/06/2025

APPENDIX 3: ABBREVIATIONS

Abbreviations	Full texts
BEF	Baseline Emission Factor
CO ₂	Carbon dioxide
CPA	Component Project Activity
EMRA	Energy Market Regulatory Authority (Republic of Türkiye)
GCC	Global Carbon Council
CL	Clarification Request
CMP	Conference of Parties Serving as Meeting of Parties
CAR	Corrective Action Request
EB	Executive Board
GS	Gold Standard
FAR	Forward Action Request
GHG	Green House Gas
MP	Monitoring Period
MR	Monitoring Report
O&M	Operation and Maintenance
PAI	Project Activity Instances
PD	Project Description
PP	Project Proponent
QA/QC	Quality Assurance and Quality Control
RCP	Renewal of Crediting Period
UNFCCC	United Nations Framework Convention on Climate Change
VCS PD	VCS Project Description
VCS	Verified Carbon Standard
VCSA	Verified Carbon Standard Association
VCUs	Verified Carbon Units
VVB	Validation and Verification Body
TEIAS	Turkish Electricity Transmission Corporation
EPIAS	Energy Markets Management Inc. (Republic of Türkiye)
QMS	Quality Management Systems

APPENDIX 4: FINDINGS OVERVIEW

Table 1. FAR from previous verification

FAR ID	01	Section No.	3.1	Date : 04/04/2025
Description of FAR				
Change in the start date of the PA has resulted in change in the duration of the first and second crediting period. The first crediting period now is from 04/10/2012 – 03/10/2022, and the second crediting period is for a subsequent 10 years. However, the VCS registry states the second monitoring period to be from 01/12/2021 to 30/11/203, which overlaps with the first crediting period. The subsequent VVB shall ensure that post issuance for 1st monitoring period, the registry webpage is updated to reflect the correct duration of the second crediting period, in line with the altered start date of PA.'				
Project participant response				Date : 12.06.2025

The Verra Registry webpage is checked, and the following statements are found for the crediting period statements:

Credit Type:	VCU
Crediting Period Type: *	10 Years Twice Renewable
Crediting Period Start: *	01/10/2012 (format DDMM/YYYY)
Crediting Period Duration Start Date:	01/10/2012 (format DDMM/YYYY)
Crediting Period Duration End Date:	30/09/2022 (format DDMM/YYYY)
Crediting Period Term:	CREDITING PERIOD HAS EXPIRED
Early Action:	No

Crediting Period Term

CREDITING PERIOD HAS EXPIRED, 01/10/2012 - 30/09/2022

So, VCS states the first crediting period as 01/10/2012 - 30/09/2022. However, the first crediting period is revised as 21/02/2014 – 20/02/2024 (both dates are included) in the approved 1st MR of the 1st CP (section 1.6). Subsequently, the second crediting period is 21/02/2024 – 20/02/2034. The description of the FAR#1 above needs to be revised accordingly.

Other than these, an email is sent to Verra for the update of the crediting periods, msg. file is shared with VVB and VVB is in the CC of the email.

Documentation provided by project participant

DOE assessment

Date: 19/06/2025

The VVB has reviewed the email sent to Verra and the snip of the Verra registry webpage showing the request by PP regarding the updating of crediting period in the Verra project portal. The VVB has checked the Verra projects portal and confirms that the crediting period has been updated. Hence the FAR has been closed.

FAR01 is closed.

Table 1.CL from this validation

CL ID	01	Section no.	3.1	Date : 30/06/2024
Description of CL				
In section 1.2 'Audit History'; Under the column 'Period', PP has mentioned the period of 1 st crediting period (CP) as "21-February-2014 to 20-February-2024". However, as per the Verra website the CP was from 01-October-2012 – 30-September-2022. PP shall clarify the inconsistency.				

Project participant response	Date : 19/03/2025
As stated in the first monitoring report (dating 8/8/2024 version 06), the start date of the project is taken as 21/02/2014 as per the commissioning date of the Umut II. The Provisional Acceptance Document dating 21/02/2014 is shared with VVB. The correction to the start date of the crediting period is done during the first verification.	
Documentation provided by project participant	
DOE assessment	Date: 04/04/2025
The assessment team reviewed the provisional acceptance certificate of the project activity and previous verification report (MP: 21/02/2014 to 20/02/2020 (including both days)) and confirms that the operational start date of the project activity is 21/02/2014 and hence the change in crediting periods.	
CL 01 is closed.	

CL ID	02	Section no.	3.3.4	Date : 04/10/2024
Description of CL				
As per para 3.9.8 of the VCS Standard v4.7, "The validity of the original baseline scenario shall be demonstrated, or where invalid, new baseline scenario shall be determined when renewing the project crediting period as follows:				
a) The validity of the original baseline scenario shall be assessed. Such assessment shall include an evaluation of the impact of new relevant national and/or sectoral policies and circumstances on the validity of the baseline scenario. b) Where it is determined that the original baseline scenario is still valid, the GHG emissions associated with the original baseline scenario shall be reassessed using the latest version of the CDM Tool to assess the validity of the original/current baseline and to update the baseline at the renewal of a crediting period. c) Where it is determined that the original baseline scenario is no longer valid, the current baseline scenario shall be established in accordance with the VCS Program rules."				
However, in section 3.4, no such information has been provided. PP shall demonstrate the validity of baseline scenario as per para 3.9.8 of the VCS standard v4.7.				
Project participant response				Date : 19/03/2025
Section 3.4 is revised. In accordance with the Tool 11 Assessment of the validity of the original/current baseline and update of the baseline at the renewal of the crediting period, Version 3.0.1, the validity of the baseline scenario is assessed step by step.				
Documentation provided by project participant				
DOE assessment				Date: 03/04/2025

The updated VCS PD has been reviewed. PP has now justified each step with evidence in accordance with the Tool 11. The justification provided has been reviewed and is found to be appropriate.

CL 02 is closed.

Table 2.CAR from this Validation (RCP)

CAR ID	01	Section no.	3.1	Date : 24/06/2024
Description of CAR				
The following observations have been made in the Cover Page of the PD: 1. The project ID of UMUT-II Hydropower Plant, Turkey when cross-checked with the Verra website is incorrect. PP shall clarify and revise the same. 2. According to VCS standard Version 4.7 Para 3.9.8 (4) - "The updated project description shall be validated in accordance with the VCS Program rules. In addition, the project shall be validated against the (current) scope of the VCS Program" 1) Therefore, PP is requested to update the Project description to the latest version available on the VCS website.				
Project participant response				Date : 19/03/2025
1. The project ID is revised as 1011 on the cover page. 2. The PD is updated to version 4.4.				
Documentation provided by project participant				
DOE assessment				Date: 04/04/2025
1. The VVB confirms that the PP has revised the 'Project ID' of UMUT-II Hydropower Plant, Turkey to 1011 as per the Verra website. Therefore, the finding stands (closed). 2. The VVB confirms that the PP has revised the version of VCS Project Standard to 4.7 and has used project description template v4.4. Therefore, the finding stands (closed).				
CAR 01 is closed.				

CAR ID	02	Section no.	3.1	Date : 14/06/2024
Description of CAR				
The following observations have been made in Section 1 of the PD:				

1. Section 1.1 'Summary Description of the Project' – The instructions per the VCS PD template v4.4 is “Provide a summary description of the project to enable an understanding of the nature of the project and its implementation, including the following (no more than one page): A summary description of the technologies/measures to be implemented by the project. The location of the project. An explanation of how the project is expected to generate GHG emission reductions or carbon dioxide removals. A brief description of the scenario existing prior to the implementation of the project. An estimate of annual average and total reductions and removals.”

PP shall include all these details and revise the section.

2)

2. Sections 1.1 and 1.6 of the PD containing information on Project Proponent - As per the Declaration letter for UMUT II provided, the authorized Project Proponent mentioned in the document is “RHG Enerturk Enerji Uretim ve Tic. A.S.” Same is mentioned on the Verra Webpage.

PD shall clarify and revise the same.

3. Section 1.1 of the PD mentions the project has estimated emission reductions of 45,855 tCO₂e. However, ER Sheet Tab ER, Cell C4, the estimated annual emission reductions are mentioned to be 32,347 tCO₂e. Please address the inconsistency and revise accordingly.
4. In section 1.2 'Audit History'; As per section 1.2 of the VCS PD template v4.4, PP shall revise the table contents mentioning the names of the DOEs which did the previous validations and verifications under the column 'Validation/verification body name'. Please revise the table as mentioned.
5. Section 1.4.1 Under General Eligibility- As per VCS PD template v4.4 section 1.4.1- “For all projects, describe and justify how the project is eligible to participate in the VCS Program. The response should: · Justify that the project activity is included under the scope of the VCS Program and not excluded under Table 2.1 of the VCS Standard. · Provide information to demonstrate that the project meets requirements related to the pipeline listing deadline, the opening meeting with the validation/verification body, and the validation deadline. · Demonstrate that the applied methodology is eligible under the VCS Program. Where applying a methodology with scale and/or capacity limits demonstrate that the project is not a fragmented part of a larger project or activity that would otherwise exceed such limits. If applicable, demonstrate that no single cluster of project activity instances exceeds the capacity limit. Include any other relevant eligibility information.” PP shall update.

3)

6. PP has omitted a section titled “*Compliance with Laws, Statutes and Other Regulatory Frameworks*” from the PD. As per the VCS PD template v4.4, section 1.15 is “*Compliance with Laws, Statutes and Other Regulatory Frameworks*”. PP shall rectify and introduce this section in the PD and change the numbers of other sections accordingly.
7. Section 1.12 'Description of the Project Activity', PP shall add more information as per the VCS PD template v4.4 which states “*Describe the project activity or activities (including the technologies or measures employed) and how it/they will achieve the GHG emission*

reductions or carbon dioxide removals. Describe the implementation schedule of project activity or activities. <i>For non-AFOLU projects:</i> <i>Include a list and the arrangement of the main manufacturing/production technologies, systems and equipment involved. Include in the description information about the age and average lifetime of the equipment based on manufacturer's specifications and industry standards, and existing and forecast installed capacities, load factors and efficiencies.</i> <i>Include the types and levels of services (normally in terms of mass or energy flows) provided by the systems and equipment that are being modified and/or installed and their relation, if any, to other manufacturing/production equipment and systems outside the project boundary. Clearly explain how the same types and levels of services provided by the project would have been provided in the baseline scenario.</i> <i>Where appropriate, provide a list of facilities, systems, and equipment in operation under the existing scenario prior to the implementation of the project."</i> PP is requested to revise as mentioned.	
Project participant response	Date : 19/03/2025
1. Section 1.1. is revised. The location, the technologies and measures implemented by the project, estimated annual average and total electricity generation and emission reductions for this crediting period are stated under Section 1.1. 2. Revised throughout the MR. 3. The total and annual emission reductions are estimated as 330,130 t CO₂ and 33,013 t CO₂/y, respectively. The ER Excel and the MR are consistent now. 4. The validation/verification body names are added under Section 1.2. 5. Section 1.4.1. is revised. The eligibility of the project according to VCS Standard v.4.7 is assessed. 6. The version of the PD is updated to 4.4 and Section 1.15 titled "Compliance with Laws, Statutes and Other Regulatory Frameworks" is present in the PD. 7. Section 1.12 is revised. The project activity including the technical details and information on how the GHG emission reductions are going to be achieved are added.	
Documentation provided by project participant	
DOE assessment	Date: 04/04/2025
1. The assessment team reviewed the revised PD and confirmed that the PP has included all the necessary information as per the VCS PD template version 4.4 instructions. Therefore, the finding stands (closed). 2. The assessment team reviewed the revised PD and is of the opinion that the PP has updated the name of the owner as 'RHG Enertürk Enerji Üretim ve Ticaret A.Ş.' wherever applicable in the PD. Therefore, the finding stands (closed).	

3. The assessment team reviewed the revised PD and confirms that the PP has rectified the estimated emissions reductions in the PD and has made it consistent with the ER sheet. Therefore, the finding stands(closed).
4. The assessment team reviewed the revised PD and confirms that the PP has included all the required details in section 1.2 'Audit History'. Therefore, the finding stands (closed).
5. As per the VCS standard v4.7, Grid-connected electricity generation activities using hydroelectric power plants are excluded in non-LDCs. And since Türkiye is a non-LDC, please reframe the justification accordingly.
6. The assessment team reviewed the revised PD and confirms that the PP has now included the omitted sections and has complied with VCS PD template v4.4. Therefore, the finding stands (closed).
7. The assessment team reviewed the revised PD and confirmed that the PP has included all the necessary information in section 1.12 as per the VCS PD template version 4.4 instructions. Therefore, the finding stands (closed).

Project participant response
Date : 29/04/2025

5. Umut II HEPP has a capacity of 24.450 MWe and therefore it is a large-scale plant. According to Table 1 of the VCS Standard v4.7, the large-scale grid connected electricity generation activities using hydroelectric power plants are excluded in LDCs:

Table 1: Excluded Project Activities

Exclusions from VCS Program Scope	Exclusion
Grid-connected electricity generation activities using hydroelectric power plants. The exclusion does not apply to ocean energy (e.g., wave, tidal, salinity gradient, and ocean thermal energy conversion). For hydro projects, large scale means a maximum capacity of greater than 15MW, where maximum capacity is the installed/rated capacity or authorized capacity (as determined in the activity approval from the project regulator, government, or similar entity), whichever is lower. Grid-connected means >50% of total generation is exported to a national or regional grid. See the VCS Program Definitions for the full definition.	Excluded in non-LDCs. Further, large-scale projects excluded in LDCs¹

And the host country Türkiye is not an LDC. So the project is not excluded.

Documentation provided by project participant
DOE assessment
Date: DD/MM/YYYY

The justification provided by the PP is insufficient because as per VCS Standard v4.7, grid-connected electricity generation activities using hydroelectric power plants are excluded in non-LDCs irrespective of the scale.

Project participant response
Date : 20/05/2025

Under Section 1.4.1 item 1, the statements regarding exclusion are deleted. And the following explanation from the registered Umut I HEPP's PD is added.

"The project is situated in a country that is not classified as a Least Developed Country (LDC) by the United Nations and is therefore not included in Table 1 of the VCS Standard version 4.7. Nevertheless, the project has been successfully registered with the Verra registry, and its activity pertains to the Renewal of Crediting Period (RCP)."

Documentation provided by project participant	
DOE assessment	Date: 27/05/2025
The VVB has assessed the PP response in revised PD and confirms that the project activity is eligible to renew the crediting period as per VCS Standard v4.7 as the project has already been registered under VCS (VCS Standard v3.0). CAR#02 is closed.	

CAR ID	03	Section no.	3.1, 3.2	Date : 01/07/2024
Description of CAR				
The following observations have been made in Section 1.18 of the PD ‘Sustainable Development Contributions’ of the PD: 1. The PD states that the total net electricity production of the project is 61,595 MWh. However, according to the ER Calculation Sheet (Tab ‘Electricity Generation’, Cell D15) provided by the PP, the net electricity generation is 722,759 MWh for this crediting period. PP shall rectify the inconsistency. 2. The PD states that the project contributes to SDG 13 with an amount of 5,656 tonnes of CO_{2e} per annum. However, according to the ER Sheet (Tab ‘ER’, Cell C4) provided by the PP, the annual estimated emission reductions is 32,347 tCO_{2e}. PP shall rectify the inconsistency. The following observations have been made under section 2 of the PD ‘Safeguards and Stakeholder engagement’ of the PD: 1. Sections 2.1.1 ‘Stakeholder Identification’ and 2.3.4 ‘Property Rights’ in the PD mention “No expropriation was required during the implementation period”. Whereas section 1.1 of the same PD mentions that “The Umut I project is in proximity to two villages which does lead to expropriation to some extent”. PD shall clarify and give explanation of this nonconformity. 2. Under the same section, PP shall fill in the following cells (as per the instructions given in the VCS PD template v4.4) where ‘Stakeholder Identification’, ‘Stakeholder diversity and changes over time’ and ‘Location of resources’ are mentioned. 3. Under the same section, PP shall provide more details in the following cell where ‘Legal or customary tenure/access rights’ is mentioned. The instructions to fill in this cell states “Describe any legal or customary tenure/access rights to territories and resources, including collective and conflicting rights, held by stakeholders, indigenous people (IPs), local communities (LCs), and customary rights holders.” PP shall update.				

4. Section 2.1.2 'Stakeholder Consultation and Ongoing Communication', PP shall fill in the following cells (as per the instructions given in the VCS PD template v4.4) where 'Date of stakeholder consultation' and 'Consultation outcome' are mentioned. 5. Under the same section, PP shall provide more details in the following cell where 'Ongoing communication' is mentioned. PP shall provide more information on the mechanisms for the ongoing communication with the stakeholders as per the VCS PD template v4.4. 6. Section 2.1.3 'Free Prior and Informed Consent' of the PD needs additional details as per the VCS PD template v4.4 such as - information relating to how the consent to implement the PA was obtained from the stakeholders. PP shall update as per the instructions given in the PD template which states <i>"Describe and demonstrate how consent to implement the project activities was obtained from those concerned, including IPs, LCs, and customary rights holders, and a transparent agreement was reached. Describe any ongoing or unresolved conflicts and demonstrate that the project does not exacerbate nor influence the outcomes of unresolved conflicts."</i> 7. Under the same section, PP shall provide more details such as- the assurance that the project has not encroached on land, relocated people without consent, and forced physical or economic displacement, in the following cell where 'Outcome of FPIC' is mentioned. PP shall add more details referring to the instructions given in the PD template v4.4. 8. Section 2.1.4 of the PD addresses 'Grievance Redress procedure' – PP shall provide additional information on the process PP will use to redress any grievance that might occur in the future, and how the process was developed with the stakeholders. PP shall refer to section 2.1.4 of the VCS PD template v4.4 for more instructions. 9. Sections 2.2 'Risks to stakeholders and the environment' and 2.4 'Ecosystem Health' of the PD needs additional justifications for every risk identified or not identified. As per the VCS PD template, justification should be provided even if no risks to stakeholders and the environment and ecosystem were identified. 10. Section 2.3.2 'Human Rights', PP shall provide more details referring to the PD template v4.4 which states <i>"Demonstrate how the project recognizes, respects, and promotes the protection of the rights of IPs, LCs, and customary rights holders in line with applicable international human rights law, and the United Nations Declaration on the Rights of Indigenous Peoples and ILO Convention 169 on Indigenous and Tribal Peoples."</i> PP shall update.	
Project participant response	Date : 19/03/2025
1. The total and the annual electricity generations are 722,759 MWh and 72,276 MWh/year. The PD and the ER are consistent now. 2. In terms of ER and EG the PD is revised. Both ER and PD are consistent now. Section 2: 1. The referred Sections (2.1.1 and 2.3.4) are revised. In accordance with the EPDK Declarations (shared with VVB), the expropriation process is ongoing. 2. The referred cells are filled. 3. The referred cell is revised. 4. The dates of stakeholder consultations are mentioned together with the consultation outcome.	

5. The communication ways between the project owner and the stakeholders are stated in cell for Ongoing Communication.
6. Section 2.1.3 is revised. Required information is added accordingly.
7. The details given in the cell where Outcome of FPIC is mentioned is in accordance with the PD template's instructions.
8. Under the development process of the referred section, redress of any grievance that might occur in the future is mentioned.
9. Under Section 2.2 and 2.4 the required justifications are stated.
10. Section 2.3.2 is revised and required justifications are stated there.

Documentation provided by project participant

DOE assessment

Date: 04/04/2025

In section 1.18

1. The assessment team reviewed the revised PD and confirms that the PP has rectified the estimated electricity generation in the PD and has made it consistent with the ER sheet. Therefore, the finding stands (closed).
2. The assessment team reviewed the revised PD and confirms that the PP has rectified the estimated emissions reductions in the PD and has made it consistent with the ER sheet. Therefore, the finding stands (closed).

In section 2,

1. The assessment team reviewed the revised PD and confirmed that the PP has rectified the non-conformity and has provided proper justification. Therefore, the finding stands (closed).
2. The assessment team reviewed the revised PD and confirmed that the PP has included all the necessary information as per the VCS PD template version 4.4 instructions, which has been verified from the EIA report. Therefore, the finding stands (closed).
3. The assessment team reviewed the revised PD and confirmed that the PP has included all the necessary information as per the VCS PD template version 4.4 instructions. Therefore, the finding stands (closed).
4. The assessment team reviewed the revised PD and confirmed that the PP has included all the necessary information as per the VCS PD template version 4.4 instructions, which has been verified from the EIA report. Therefore, the finding stands (closed).
5. The assessment team reviewed the revised PD and confirmed that the PP has included all the necessary information as per the VCS PD template version 4.4 instructions. Therefore, the finding stands (closed).
6. The assessment team reviewed the revised PD and confirmed that the PP has included all the necessary information as per the VCS PD template version 4.4 instructions, which has been verified from the EIA report. Therefore, the finding stands (closed).
7. The assessment team reviewed the revised PD and confirmed that the PP has included all the necessary information as per the VCS PD template version 4.4 instructions. Therefore, the finding stands (closed).

8. The assessment team reviewed the revised PD and confirmed that the PP has included all the necessary information as per the VCS PD template version 4.4 instructions and has provided a grievance book at site. Therefore, the finding stands (closed).
 9. The assessment team reviewed the revised PD and confirmed that the PP has included all the necessary information as per the VCS PD template version 4.4 instructions and has provided proper justifications. Therefore, the finding stands (closed).
 10. The assessment team reviewed the revised PD and confirmed that the PP has included all the necessary information as per the VCS PD template version 4.4 instructions. Therefore, the finding stands (closed).
- 4)

CAR 03 is closed.

CAR ID	04	Section no.	3.3.6	Date :	02/07/2024
Description of CAR					
Following observations were made under section 4 of the Project description (PD):					
1. Section 4.1 'Baseline Emissions', the equation used to calculate the value of "Combined margin CO₂ emission factor for grid connected power generation in year y ($EF_{grid,CM,y}$)" is incorrect. PP shall refer to paragraph 86 (b) of section 6.6.1 of TOOL07 v7.0 and revise. 2. Section 4.2 'Project Emissions', PP shall provide more details referring to the PD template v4.4 which states "<i>Describe the procedure for quantification of project emissions and/or carbon stock changes in accordance with the applied methodology. Project emissions may be negative where carbon stock increases (sinks) exceed project emissions. Specify the reductions and removals separately where the applied methodology provides procedures and equations to do so. Include all relevant equations here and provide sufficient information to allow the reader to reproduce the calculations. Explain and justify all relevant methodological choices (e.g., with respect to selection of emission factors and default values). Include all calculations in the emission reduction and removal calculation spreadsheet.</i>" PP shall include the calculation of power density using the values applicable to this PA. Also, state whether the value of power density is greater than 10 W/m² and what it implies. 3. Section 4.4 'Estimated GHG Emission Reductions and Carbon Dioxide Removals', PP shall include more details as per the VCS PD template v4.4 guidelines, "<i>Describe the procedure for the quantification of estimated GHG emission reductions (reductions) and carbon dioxide removals (removals). Include all relevant equations.</i>" <i>For data and parameters monitored, use the estimated data/parameter values provided in Section 5.2 below. Document how each equation is applied in a manner that enables the reader to reproduce the calculations. Provide calculations for all key equations to allow the reader to reproduce the annual calculations for estimated reductions or removals. Specify the reductions and removals separately where the applied methodology provides procedures and equations to do so. Include all of the above in the emission reduction and removal calculation spreadsheet.</i>					
Project participant response					Date :
Section 4:					19/03/2025
1. In accordance with Tool 07 paragraph 85 and 86 the equation is revised under Section 4.1.					

2. The required details are given under Section 4.2 now. PD calculation is given as well.
3. The ER calculation as per the applied methodology is given under Section 4.4. Under Section 5.2 also the equations used to calculate parameter values are addressed.

Documentation provided by project participant

DOE assessment

Date: 04/04/2025

1. The assessment team reviewed the revised PD and confirms that the PP has referred to paragraph 86 (b) of section 6.6.1 of TOOL07 v7.0 and has used the applicable equation to calculate the value of “Combined margin CO₂ emission factor for grid connected power generation in year y ($EF_{grid,CM,y}$). Therefore, the finding stands (closed).
2. The assessment team reviewed the revised PD and confirms that the PP has included all the necessary calculations and justifications in section 4.2 of the PD which has been verified from the generation license and area of the reservoir calculated by a third party. Therefore, the finding stands (closed).
3. The assessment team reviewed the revised PD and confirmed that the PP has included all the necessary information as per the VCS PD template version 4.4 instructions, as verified from the ER calculation sheet. Therefore, the finding stands (closed).

CAR 04 is closed.

CAR ID	05	Section no.	3.3.8	Date : 14/06/2024
Description of CAR				
Section 5.1 of the PD ‘Data and Parameters Available at Validation’, PP shall provide more details in the following cell where ‘Justification of choice of data or description of measurement methods and procedures applied’ is mentioned. The instructions to fill in this cell as per the PD template v4.4 are <i>“Justify the choice of data source, providing references where applicable. Where values are based on measurement, include a description of the measurement methods and procedures applied (e.g., what standards or protocols have been followed), indicate the responsible person/entity that undertook the measurement, the date of the measurement and the measurement results. More detailed information may be provided in an appendix.”</i> PP shall update. Following observations were made under section 5.2 ‘Data and Parameters Monitored’ of the Project description (PD): 1. PP shall provide more details in the following cell where ‘Description of measurement methods and procedures to be applied’ is mentioned. The instructions to fill in this cell as per the PD template v4.4 are <i>“Specify the measurement methods and procedures, any standards or protocols to be followed, and the person/entity responsible for the measurement. Include any relevant information regarding the accuracy of the measurements (e.g., accuracy associated with meter equipment or laboratory tests).”</i> PP shall update. 2. PP shall also provide the calibration details of both the meters along the frequency of calibration under QA/QC procedures. 3. Under the same parameter mentioned above, in the following cell where ‘Calculation method’ is mentioned, the installed capacities of UMUT-I and UMUT-III are inconsistent with the values given in the production license of the project. PP is requested to revise and recalculate the values of “Generation of UMUT-I H YDROPOWER PLANT, TURKEY” and “UMUT-III HYDROPOWER PLANT, TURKEY”.				
Project participant response				Date : 19/03/2025
Under Section 5.1, more details regarding combined margin emission factor are given in the cell where “Justification of choice of data or description of measurement methods and procedures applied” is mentioned. Section 5.2: 1. The details related to the measurement methods are specified in the referred cell for EG parameter. 2. Details regarding the calibration of the meters are mentioned in QA/QC procedures now. 3. The installed capacities are given as follows in the generation license.				

Ünite kurulu güçleri : (2x3.037)+(3x8,435)+(3x4,255)MWm/ (2x2.900)+(3x 8,150)+(3x4,000) MW _e	
Here, Umut I, II and III have capacities of 2*2.9, 3*8.15 and 3*4, respectively. Both ER and PD is checked in terms of corresponding installed capacities, the values are correct.	
Documentation provided by project participant	
DOE assessment	
Date: 04/04/2025	
The assessment team reviewed the revised PD and confirmed that the PP has included all the necessary information as per the VCS PD template version 4.4 instructions which has been verified from Turkey's National Electricity Network Emission Factor Factsheet for 2022 published on 26-November 2024. Therefore, the finding stands (closed).	
Section 5.2	
1. PP is requested add more details as per the instructions given above. 2. The assessment team reviewed the revised PD and confirmed that the PP has included all the necessary information as per the VCS PD template version 4.4 instructions, as verified from the Measurement and Measuring Instruments Inspection Regulation dated 24/07/1994 and the calibration certificates provided by the PP. Therefore, the finding stands (closed). 3. The assessment team revied the revised PD and confirms that the PP has rectified the installed capacities of UMUT-I and UMUT-II in the PD and has made it consistent with generation license and has been cross-verified from the provisional acceptance certificates. Therefore, the finding stands(closed).	
Project participant response	
Date : 29/04/2025	
1. More details regarding description of measurement methods for EG parameter are added.	
Documentation provided by project participant	
DOE assessment	
Date: 05/05/2025	
The VVB has assessed the revised PD and confirms that the PP has provided all necessary information regarding the measurement methods and procedures of the parameter "EG _{PJ, facility, y} " which has been verified from the EPIAS and TEIAS websites and EPIAS invoices. Therefore, the finding stands closed.	
CAR#05 is closed.	

CAR ID	06	Section no.	3.3.8	Date : 02/07/2024
Description of CAR				
Section 5.3 of the PD ‘Monitoring Plan’, PP shall provide more details referring to the VCS PD template v4.4 such as - the procedures for calibrating monitoring equipment. The instructions to fill in this section are as below <i>“Describe the process and schedule for obtaining, recording, compiling and analyzing the monitored data and parameters set out in Section 5.2 (Data and Parameters Monitored) above.</i> <i>Include details on the following:</i> <i>• The methods for measuring, recording, storing, aggregating, collating and reporting on monitored data and parameters. Where relevant, include the procedures for calibrating monitoring equipment.</i> <i>• The organizational structure, responsibilities and competencies of the personnel that will be carrying out monitoring activities.</i> <i>• The procedures for internal auditing and QA/QC.</i> <i>• The procedures for handling non-conformances with the validated monitoring plan.</i> <i>• Any sampling approaches used, including target precision levels, sample sizes, sample site locations, stratification, frequency of measurement and QA/QC procedures.</i> <i>Where appropriate, include line diagrams to display the GHG data collection and management system.”</i> PP is requested to update as per the instructions stated above.				
Project participant response				Date : 19/03/2025
Under Section 5.3 of the PD, the calibration procedure of the meters is mentioned. The required details regarding monitoring parameters are stated.				
Documentation provided by project participant				
DOE assessment				Date: 04/04/2025
The assessment team reviewed the revised PD and confirmed that the PP has included all the necessary information as per the VCS PD template version 4.4 instructions, as verified from the previous verification report (version 2.0 dated 08/08/2024) and the calibration certificates provided by the PP. Therefore, the finding stands (closed). CAR 06 is closed.				

Table 3.FAR from this validation

FAR ID		Section No.		Date : 17/06/2025
Description of FAR				

<i>There is no FAR from this validation</i>	
Project participant response	Date : DD/MM/YYYY
Documentation provided by project participant	
DOE assessment	Date: DD/MM/YYYY

APPENDIX 5: COMPETENCY STATEMENTS

Competence Statement (ISO 14065)			
Name	Divij Varshney		
Education	M. Tech. Renewable energy systems B.Tech. Electrical Engineering		
Experience	3.5 years		
Field	Climate Change & Environment / Industry		
Approved Roles			
Team Leader	Yes (VM)		
Validator	Yes (VM)		
Verifier	Yes (VM)		
Local expert	Yes (India)		
Financial Expert	NO		
Technical Reviewer	Yes		
TA Expert (X.X)	Yes (1.2 &3.1)		
Reviewed by	Shifali Guleria, Quality Manager	Date	04/04/2024
Approved by	Deepika Mahala, Technical Manager	Date	04/04/2024

Competence Statement			
Name	Diyotima Karfa		
Education	B.Tech. Biotechnology M.Sc. Biotechnology		
Experience	1.5 year		
Field	Climate Change		
Approved Roles			
Team Leader	YES		
Validator	YES		
Verifier	YES		
Local expert	YES (India)		
Financial Expert	NO		
Technical Reviewer	NO		
TA Expert (X.X)	YES (TA 1.2, 3.1)		
Reviewed by	Shifali Guleria (Quality Manager)	Date	17/02/2025
Approved by	Deepika Mahala (Technical Manager)	Date	17/02/2025

Competence Statement			
Name	Austin Paul		
Education	B.Sc Geology - Geochemistry, Petrology, Palaeontology M.Sc Environmental Management - EIA, Environmental Policy		
Experience	NA		
Field	Environmental Management, Climate change		
Approved Roles			
Team Leader	NO		
Validator	Yes		
Verifier	Yes		
Methodology Expert	NO		
Local expert	NO		
Financial Expert	NO		
Technical Reviewer	NO		
TA Expert (X.X)	NO		
Reviewed by	Shifali Guleria (Quality Manager)	Date	21-10-2024
Approved by	Deepika Mahala (Technical Manager)	Date	21-10-2024

Competence Statement	
Name	Kubra Agriman
Education	BS Environmental Engineering
Experience	2 years
Field	Environmental Engineering
Approved Roles	
Team Leader	NO
Validator	NO
Verifier	NO
Methodology Expert	NO
Local expert	Yes (Türkiye)
Financial Expert	NO
Technical Reviewer	NO
TA Expert (X.X)	NO
Trainee	Yes

Reviewed by	Shifali Guleria, Quality Manager	Date	27/12/2022
Approved by	Deepika Mahala, Technical Manager	Date	27/12/2022

Competence Statement			
Name	Jinesh Amlani		
Education	M.Sc. Energy Systems B.Sc. Physics		
Experience	8+ years		
Field	Climate Change & Environment		
Approved Roles			
Team Leader	YES (VM Only)		
Validator	YES		
Verifier	YES		
Local expert	YES (India)		
Financial Expert	YES (VM Only)		
Technical Reviewer	YES		
TA Expert (X.X)	YES (TA.1.2, TA 3.1)		
Reviewed by	Shifali Guleria (Quality Manager)	Date	21/03/2025
Approved by	Deepika Mahala (Technical Manager)	Date	21/03/2025